

The slide features abstract geometric shapes in the corners. The top-left corner is filled with overlapping triangles in shades of blue, green, and red. The bottom-right corner contains a cluster of overlapping triangles in various shades of gray.

CHAPTER 1

REACTION KINETICS



1.1 Reaction Rate

REACTION KINETICS

- **Reaction kinetics** is the study of the rates of chemical reactions, the factors that affect these rates and the reaction mechanisms by which reactions occur.

RATE OF REACTION

- The **reaction rate** or **rate of reaction** is the speed at which a chemical reaction takes place, defined as proportional to the increase in the concentration of a product per unit time and to the decrease in the concentration of a reactant per unit time.

- **Reaction rate** is the change in the concentration of a reactant or a product with time.
- Unit of rate ($\text{mol L}^{-1} \text{time}^{-1}$).

$$\text{Rate} \propto \frac{1}{\text{time}}$$

Example, $A \rightarrow B$ * dt = period of time

$$\text{Rate} = - \frac{d[A]}{dt} \quad d[A] = \text{change in concentration of A}$$

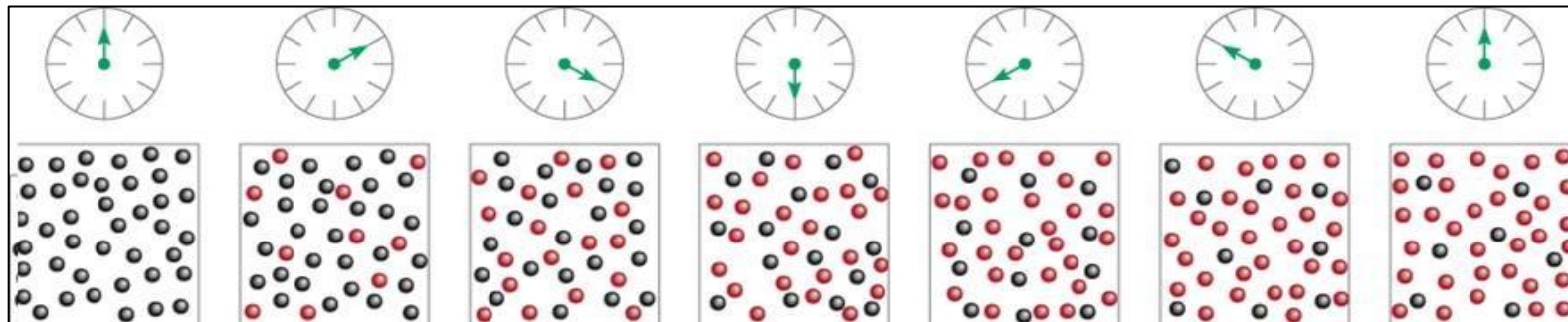
$$\text{Rate} = + \frac{d[B]}{dt} \quad d[B] = \text{change in concentration of B}$$

- ❑ Because $[A]$ decreases with time, $d[A]$ is negative.
- ❑ Because $[B]$ increases with time, $d[B]$ is positive.



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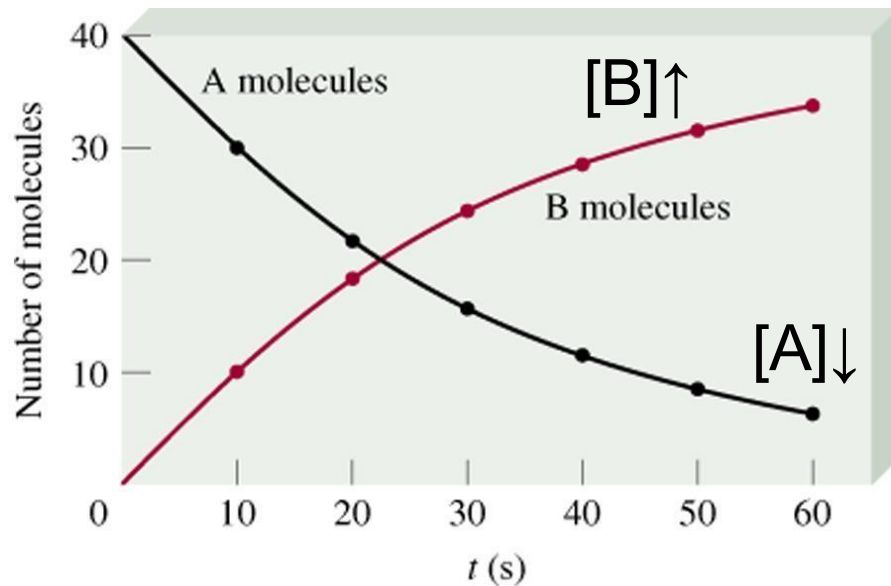
Reaction
progression



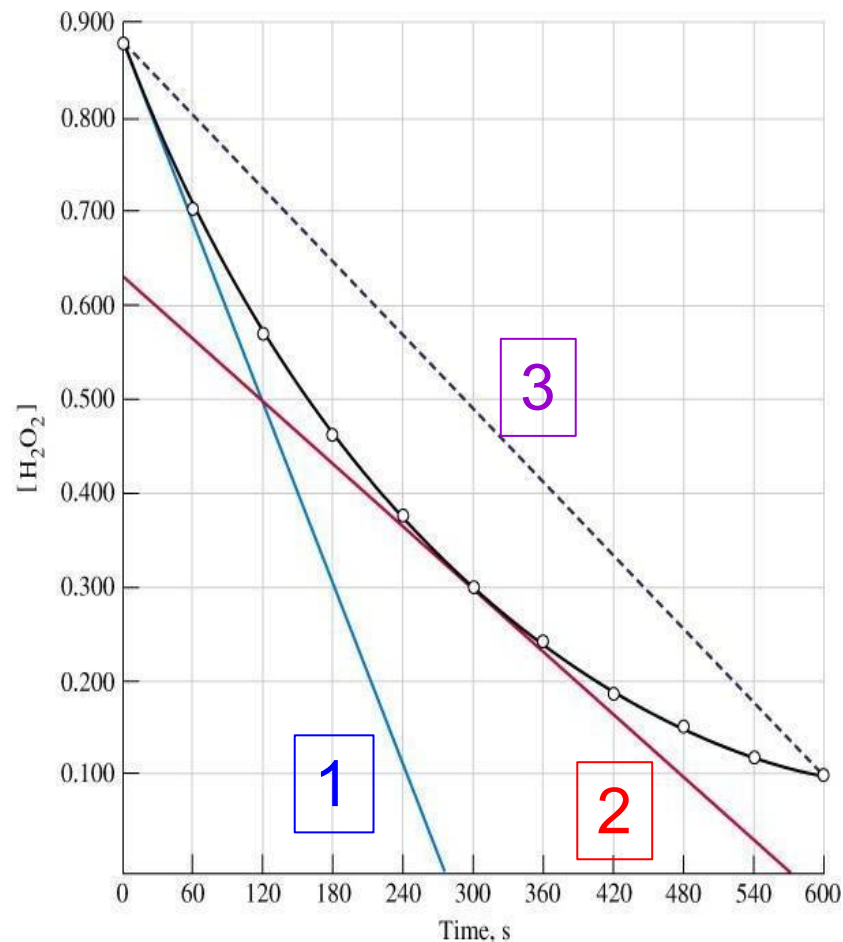
Time

$$\text{Rate} = -\frac{d[A]}{dt}$$

$$\text{Rate} = \frac{d[B]}{dt}$$



- The **average rate** is the rate over a period of time. The rate of reaction at a given time is called the **instantaneous rate** of reaction.
- The instantaneous rate at the beginning of a reaction is called **initial rate**.
- Instantaneous rate is determined from a graph of concentration vs time by drawing a tangent of the curve at that particular time.



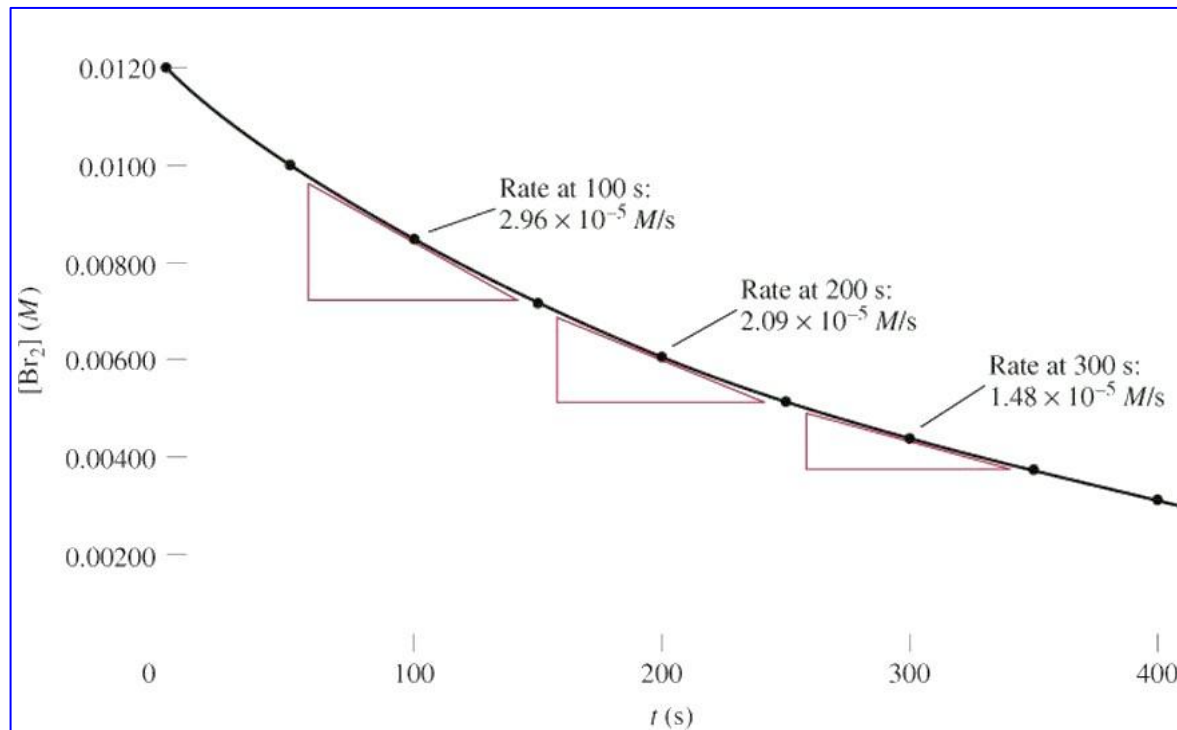
For reaction,
 $H_2O_2(aq) \rightarrow H_2O(l) + \frac{1}{2} O_2(g)$

Reaction rates are obtained from the slopes of the straight lines;

- **Initial rate** (1).
- **Instantaneous rate** at $t = 300$ s (2).
- **Average rate** (3).



Time (s)	[Br ₂] (M)
0.0	0.0120
50.0	0.0101
100.0	0.00846
150.0	0.00710
200.0	0.00596
250.0	0.00500
300.0	0.00420
350.0	0.00353
400.0	0.00296

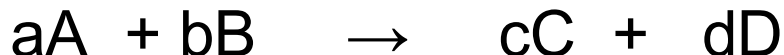


$$\text{Average rate} = - \frac{d[\text{Br}_2]}{dt} = - \frac{[\text{Br}_2]_{\text{final}} - [\text{Br}_2]_{\text{initial}}}{t_{\text{final}} - t_{\text{initial}}}$$

Differential Rate Equation

- **Differential rate equation** represents the **relationship between the rate of disappearance of reactants and rate of formation of product.**

Consider the reaction,



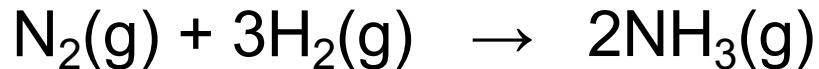
$$\text{Rate} = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = +\frac{1}{c} \frac{d[C]}{dt} = +\frac{1}{d} \frac{d[D]}{dt}$$

a, b, c and d are the stoichiometric coefficients.

Differential Rate Equation

Example:

Consider the following equation,



The differential rate equation is;

$$\text{Rate} = -\frac{d[\text{N}_2]}{dt} = -\frac{1}{3} \frac{d[\text{H}_2]}{dt} = +\frac{1}{2} \frac{d[\text{NH}_3]}{dt}$$

The equation means that the **rate of disappearance of N_2** is **$1/3$ the rate of disappearance of H_2** and **$1/2$ the rate of formation of NH_3** .

Example:

Consider the reaction, $2\text{HI} \rightarrow \text{H}_2 + \text{I}_2$, determine the rate of disappearance of HI if the rate of I_2 formation is $1.8 \times 10^{-6} \text{ Ms}^{-1}$.

Solution:

$$\text{Rate} = -\frac{1}{2} \frac{d[\text{HI}]}{dt} = +\frac{d[\text{H}_2]}{dt} = +\frac{d[\text{I}_2]}{dt}$$

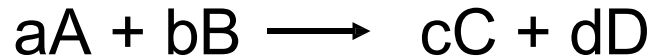
$$\begin{aligned} \frac{d[\text{I}_2]}{dt} &= 1.8 \times 10^{-6} \\ + \frac{d[\text{I}_2]}{dt} \end{aligned}$$

$$\text{Rate} = -\frac{1}{2} \frac{d[\text{HI}]}{dt} = +\frac{d[\text{I}_2]}{dt}$$

$$-\frac{d[\text{HI}]}{dt} = 2 \times 1.8 \times 10^{-6} = \mathbf{3.6 \times 10^{-6} \text{ Ms}^{-1}}$$

Rate Law

- **Rate law** expresses the relationship of the rate of a reaction to the rate constant and the concentrations of the reactants raised to some powers.



$$\text{Rate} = k [A]^x[B]^y$$

- Reaction order is x with respect to A.
- Reaction order is y with respect to B.
- Overall reaction order is $(x + y)$.
- The exponents x , y can be integers, fractions or decimal or negative values.
- k is called rate constant.

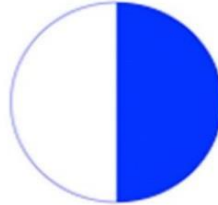
Half-life

- Half-life ($t_{1/2}$)** is the time required for the concentration (or amount) of a reactant to reach half of its initial value.

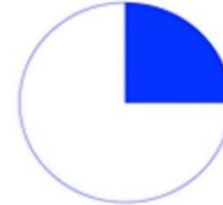
Initial amount
100%



After 1st half-life,
50% remains



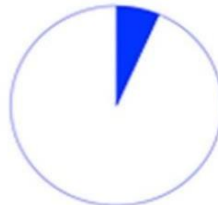
After 2nd half-life,
25% remains



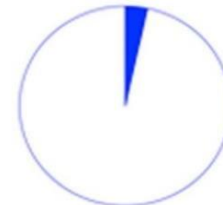
After 3rd half-life,
~12% remains



After 4th half-life,
~6% remains



After 5th half-life,
~3% remains



Reaction order

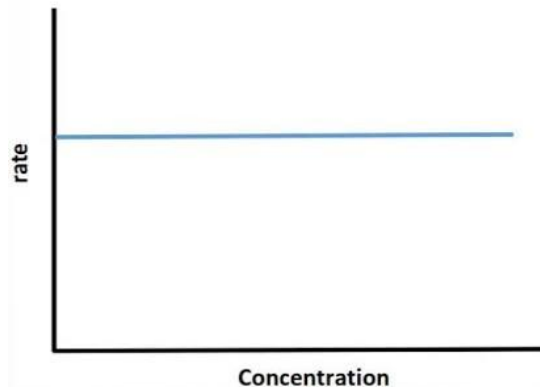
- **Order of reaction** is the power to which the concentration of a reactant is raised in the rate law.
- For example, the rate of a first-order reaction is dependent solely on the concentration of one species in the reaction.

- Some characteristics reaction order are listed below:
 1. Reaction order represents the species whose concentration **directly affects the rate of reaction**.
 2. It can be obtained by adding all the exponents of the concentration terms in the rate expression/rate law.
 3. The order of reaction **does not depend** on the stoichiometric coefficients.
 4. The reaction order of a chemical reaction **defined by reactant concentrations** and **not by product concentrations**.
 5. The order of reaction can be **zero, an integer or a fraction**.

Reaction Order & Dependence of Reaction Rate on the Concentration of Reactant

Consider a general reaction: $A \rightarrow \text{product}$

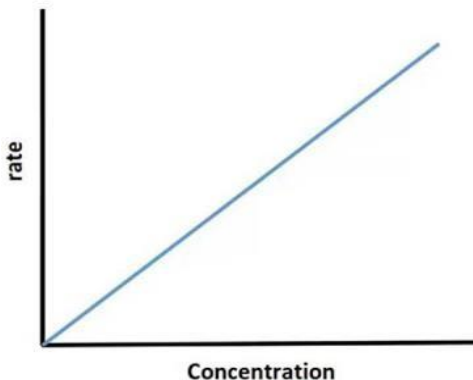
Zero order



$$\text{Rate} = k [A]^0$$

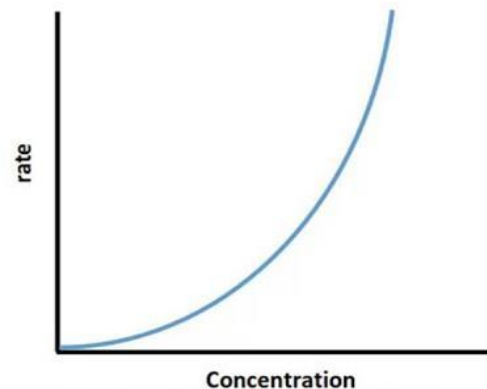
Rate **independent**
of $[A]$

First order



$$\text{Rate} = k [A]^1$$

Second order

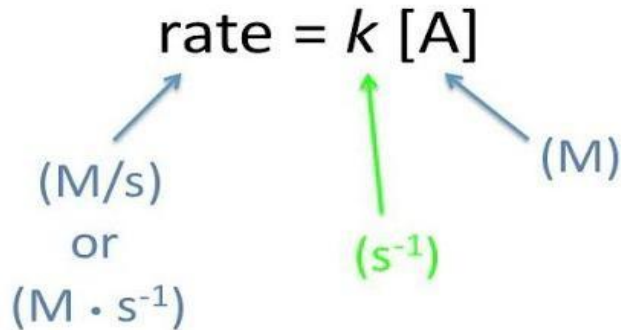


$$\text{Rate} = k [A]^2$$

Rate **dependent** on $[A]$

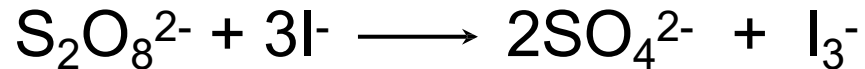
Reaction Order & Unit of Rate Constants (**k**)

- The units of **rate constant, k**, depends on the **order of reaction** and can be determined from the **rate law**.
- Example: For first-order reaction



Reaction order	Units for k	Units for Rate
0	$\frac{M}{s}$ or $M \cdot s^{-1}$	$\frac{M}{s}$ or $M \cdot s^{-1}$
1	$\frac{1}{s}$ or s^{-1}	$\frac{M}{s}$ or $M \cdot s^{-1}$
2	$\frac{1}{M \cdot s}$ or $M^{-1} \cdot s^{-1}$	$\frac{M}{s}$ or $M \cdot s^{-1}$

Example:



The above reaction is first-order with respect to the iodide ion and thiosulfate ion.

a) Write the rate equation for the reaction.

b) What is the unit of rate constant, k ?

Solution:

a) $\text{Rate} = k [\text{S}_2\text{O}_8^{2-}]^1 [\text{I}^-]^1$

b)
$$k = \frac{\text{Rate}}{[\text{S}_2\text{O}_8^{2-}]^1 [\text{I}^-]^1}$$

$$= \frac{\text{Ms}^{-1}}{\text{M}^2} = \text{M}^{-1}\text{s}^{-1}$$

Integrated Rate Equation

- Rate laws can be converted into equations that **relate the concentrations of reactants to time** during the course of a reaction.
- The resulting equations are known as an **integrated rate equations**.
- Each type of reaction order has its typical or very own integrated rate equation.

The **Integrated Rate Equation** for **Zero-order** Reaction

- The reaction rate is **independent** to the reactant concentration.
- Consider a zero-order reaction: $A \rightarrow \text{product}$

$$\text{Rate} = k [A]^0; \quad \text{Rate} = k; \quad \frac{d[A]}{dt} = -k$$

- The integrated rate equation for a zero-order reaction is:

$$[A]_0 - [A]_t = kt$$

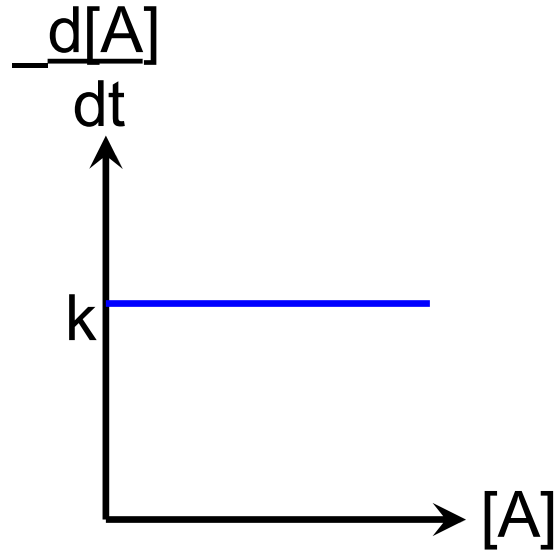
$[A]_0$ = initial concentration of reactant A.

$[A]_t$ = concentration of A that is remained after t time.

Example of graphs of a zero-order reaction:

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Rate vs concentration

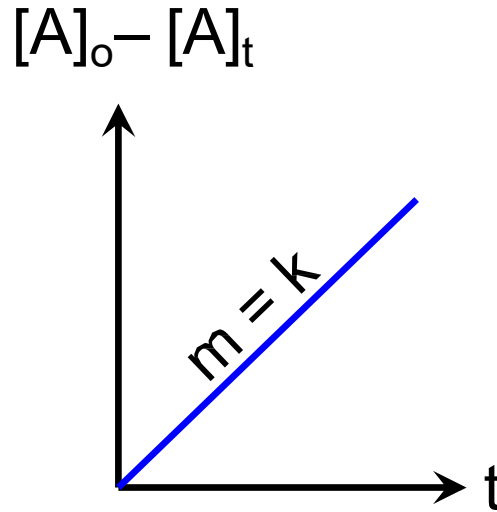


$$-\frac{d[A]}{dt} = k$$

$y = mx + c$

Dashed arrows indicate the mapping from the equation above to the y and c terms in the general linear equation below.

$[A]_o - [A]_t$ vs time, t

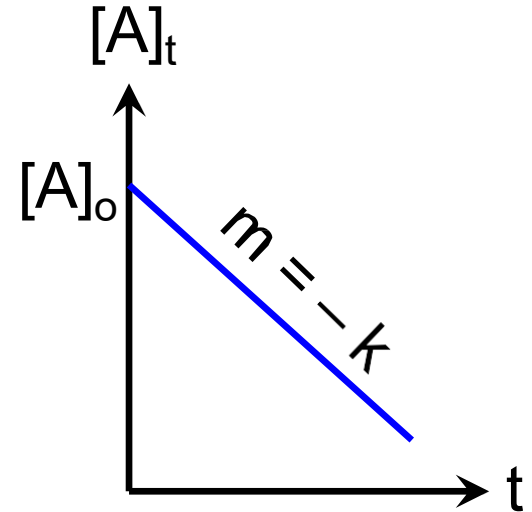


$$[A]_o - [A]_t = kt$$

$y = mx + c$

Dashed arrows indicate the mapping from the equation above to the y , m , and c terms in the general linear equation below.

Concentration vs t



$$[A]_t = -kt + [A]_o$$

$y = mx + c$

Dashed arrows indicate the mapping from the equation above to the y , m , and c terms in the general linear equation below.

The **Half-life** of Zero-order Reaction

$$\text{At } t = t_{1/2}, [A]_t = \frac{[A]_o}{2}$$

Substituting $[A]_t = \frac{[A]_o}{2}$ into the integrated rate law,

$$[A]_o - [A]_t = kt$$

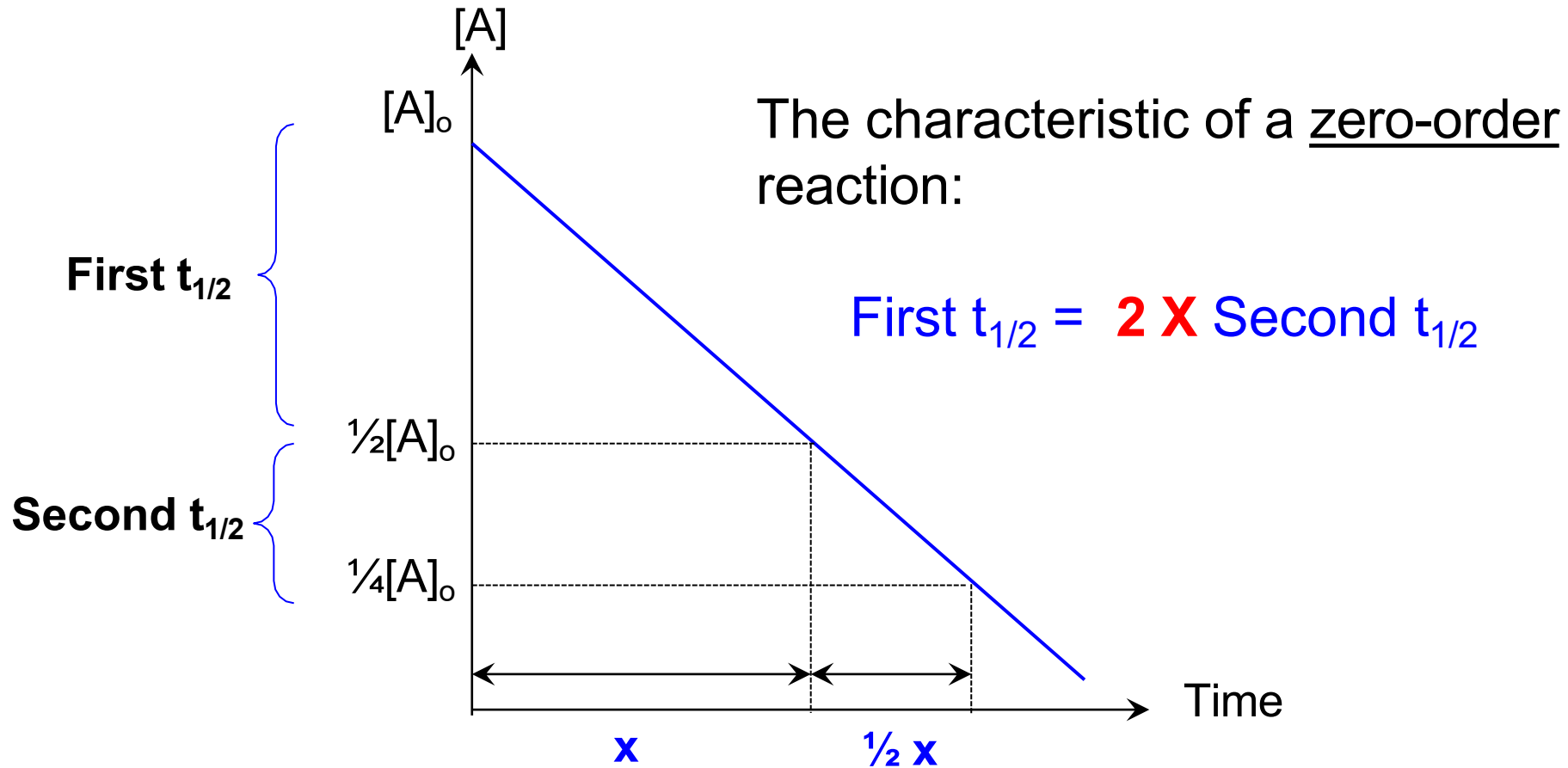
$$[A]_o - \frac{[A]_o}{2} = kt_{1/2}$$

$$t_{1/2} = \frac{[A]_o}{2k}$$

Note :

The half-life of a zero-order reaction is directly proportional to $[A]_o$.

The **Half-life** of Zero-order Reaction



The **Integrated Rate Equation** for First-order Reaction

- The reaction rate is **directly proportional** to the reactant concentration.
- Consider a first-order reaction: $A \rightarrow \text{product}$

$$\text{Rate} = k [A]^1; \quad -\frac{d[A]}{dt} = k[A]$$

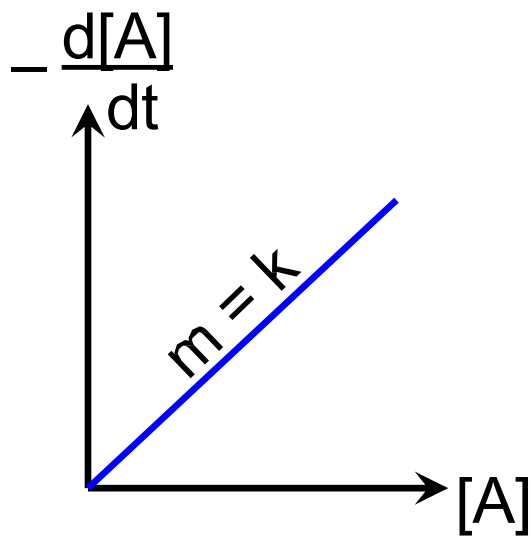
- The integrated rate equation for a first-order reaction is:

$$\ln [A]_0 - \ln [A]_t = kt$$

@

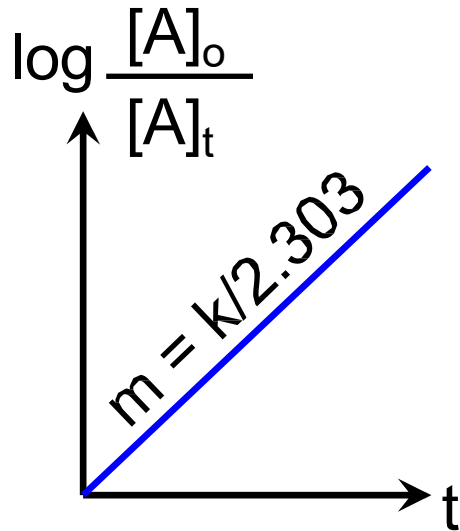
$$\log [A]_0 - \log [A]_t = \frac{kt}{2.303}$$

Example of graphs of a first-order reaction:



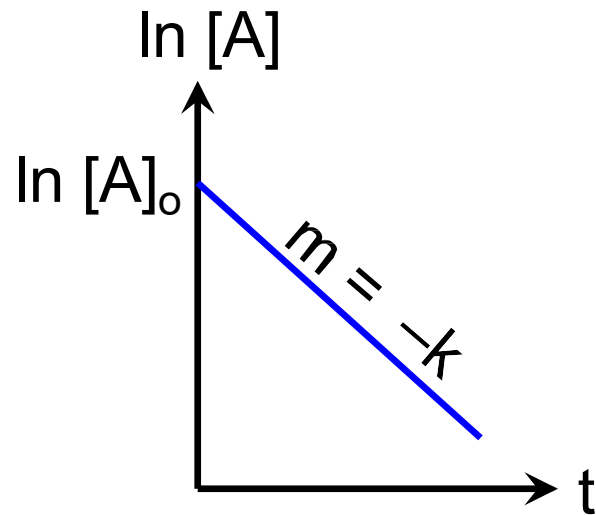
$$-\frac{d[A]}{dt} = k[A]$$

Below the equation, a linear form is shown: $y = mx + c$. Dashed arrows indicate the mapping: y points to $-\frac{d[A]}{dt}$, x points to $[A]$, m points to k , and c points to 0 .



$$\log \frac{[A]_0}{[A]_t} = \frac{k}{2.303} t$$

Below the equation, a linear form is shown: $y = mx + c$. Dashed arrows indicate the mapping: y points to $\log \frac{[A]_0}{[A]_t}$, x points to t , m points to $k/2.303$, and c points to 0 . The fraction $k/2.303$ is circled in the original image.



$$\ln [A]_t = \ln [A]_0 - kt$$

Below the equation, a linear form is shown: $y = mx + c$. Dashed arrows indicate the mapping: y points to $\ln [A]_t$, x points to t , m points to $-k$, and c points to $\ln [A]_0$.

The **Half-life** of First-order Reaction

$$\text{At } t = t_{1/2}, [A]_t = \frac{[A]_o}{2}$$

Substituting $[A]_t = \frac{[A]_o}{2}$ into the integrated rate law,

$$\ln [A]_o - \ln [A]_t = kt$$

$$\ln [A]_o - \ln \frac{[A]_o}{2} = kt$$

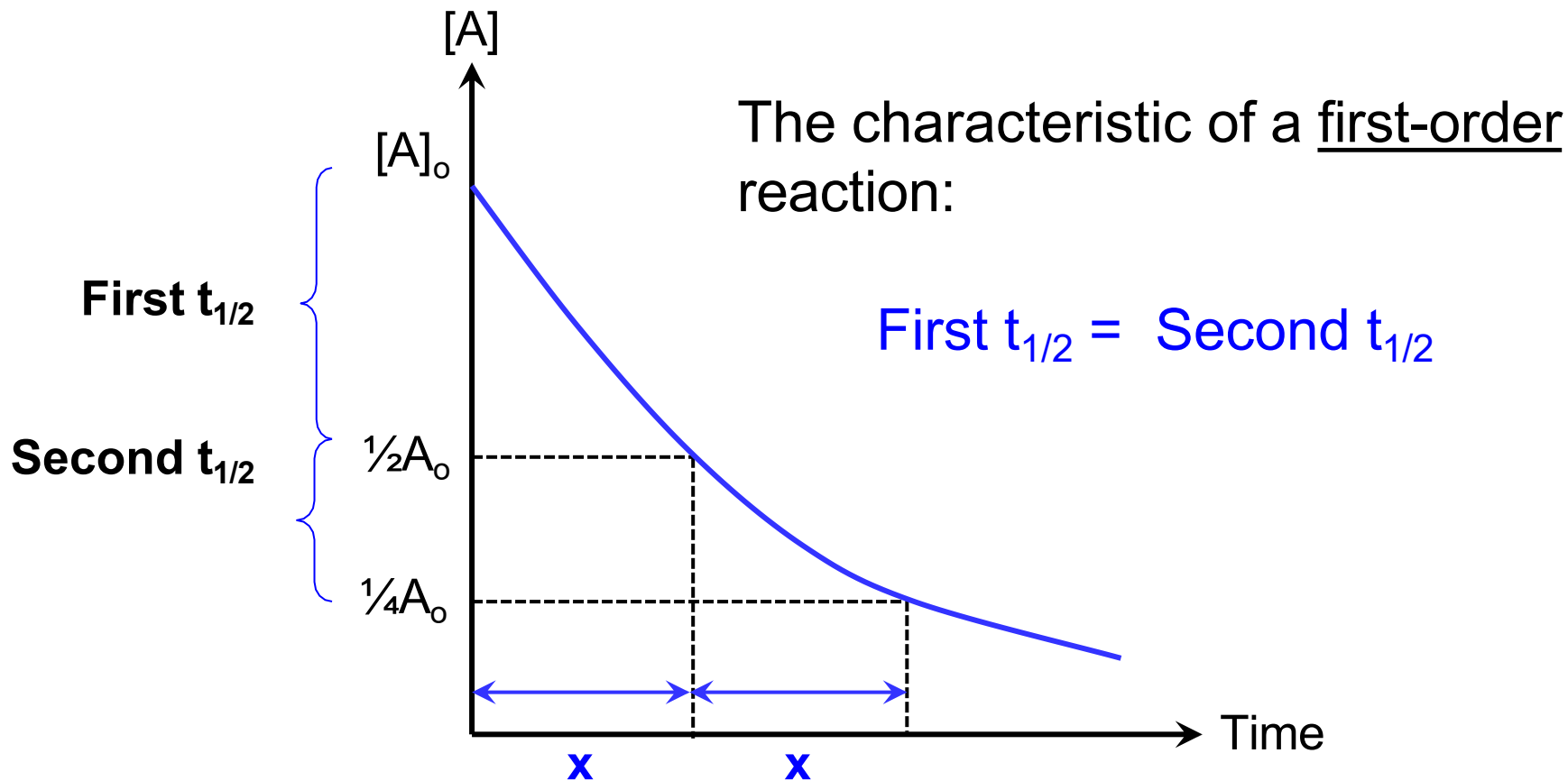
$$\ln \frac{[A]_o}{[A]_o/2} = kt_{1/2}$$

$$t_{1/2} = \frac{\ln 2}{k}$$

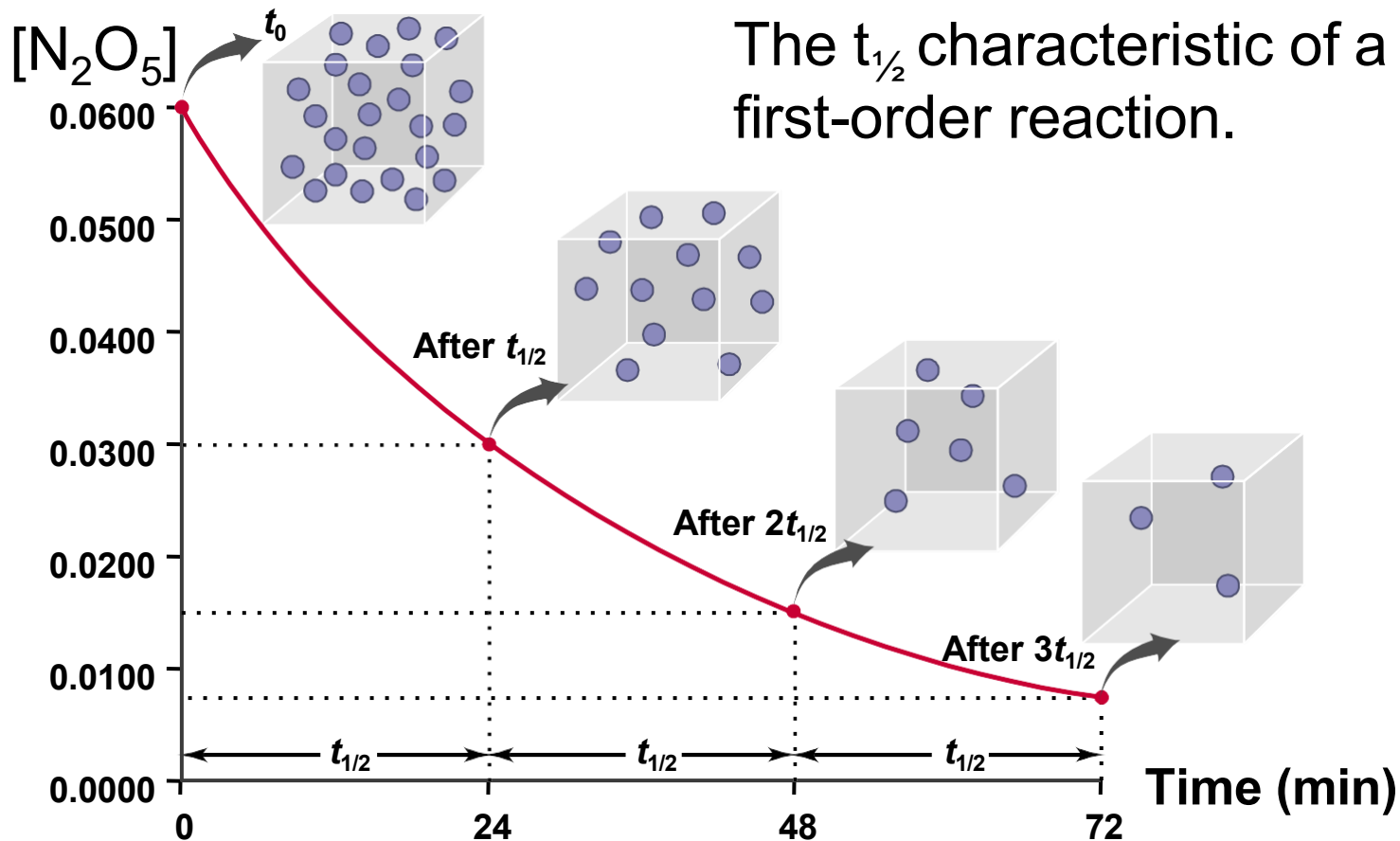
Note :

The half-life of a first order reaction is constant and independent of $[A]_o$.

The **Half-life** of First-order Reaction



Example:



The **Integrated Rate Equation** for Second-order Reaction²⁹

- The reaction rate is **directly proportional** to a **single reactant concentration** raised to the power of two.
- Consider a second-order reaction: $A \rightarrow \text{product}$

$$\text{Rate} = k[A]^2;$$

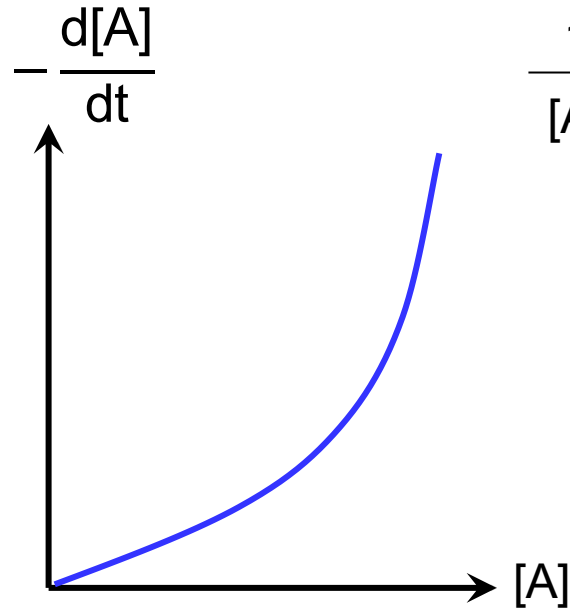
$$\frac{d[A]_o}{dt} = k[A]^2$$

- The integrated rate equation for a second-order reaction:

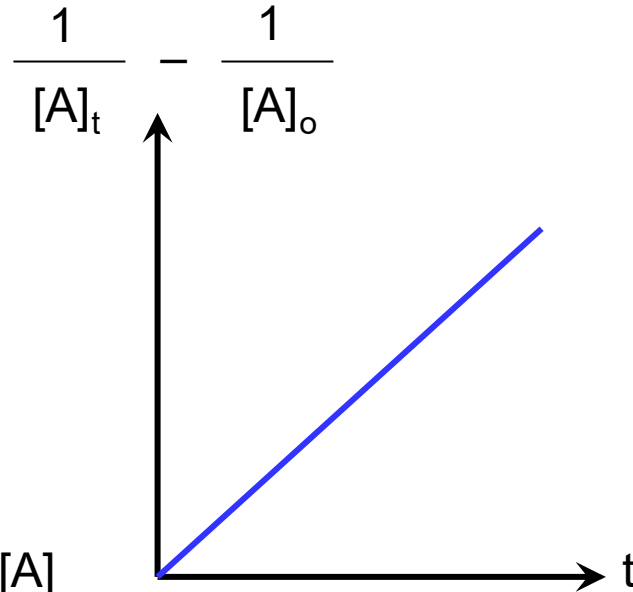
$$\frac{1}{[A]_t} - \frac{1}{[A]_o} = kt$$

Example of graphs of a second order reaction:

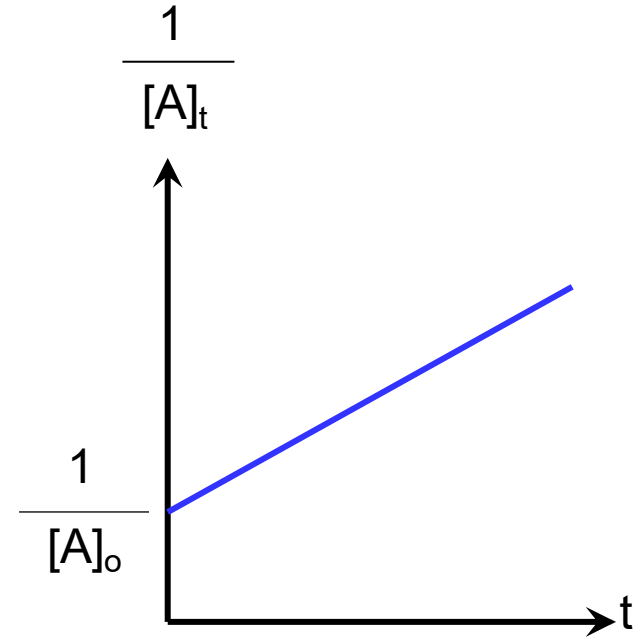
Rate vs concentration



$1/[A]_t - 1/[A]_o$ vs time, t



$1/[A]_t$ vs time, t



The **Half-life** of Second-order Reaction

$$\text{At } t = t_{1/2}, [A]_t = \frac{[A]_o}{2}$$

Substituting $[A]_t = \frac{[A]_o}{2}$ into the integrated rate law,

$$\frac{1}{[A]_t} - \frac{1}{[A]_o} = kt \quad t_{1/2} = \frac{1}{k[A]_o}$$

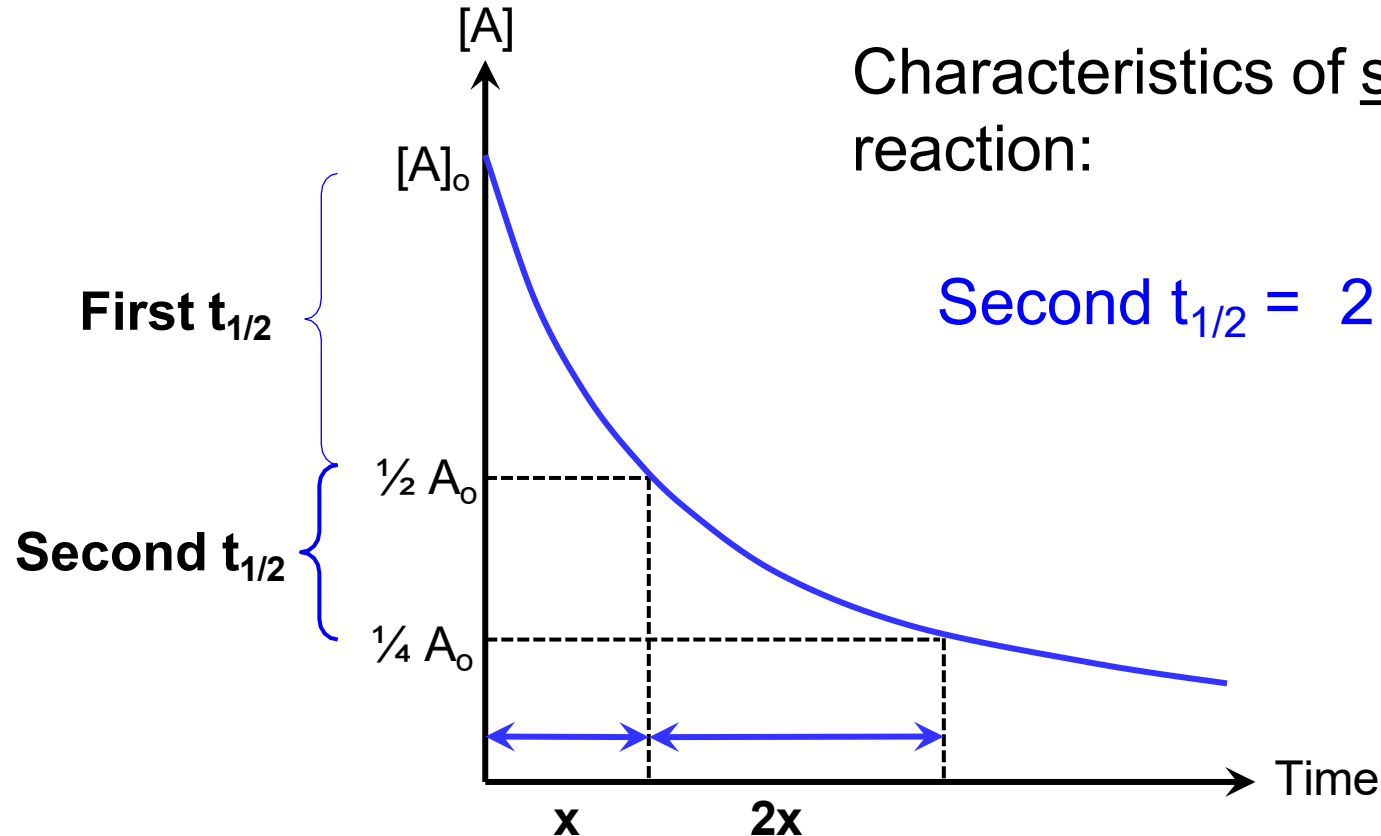
Note :

The half-life of a second-order reaction is **inversely proportional** to $[A]_o$.

The **Half-life** of Second-order Reaction

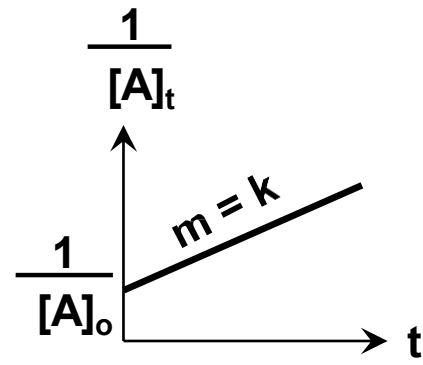
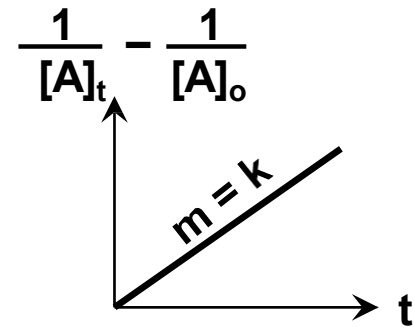
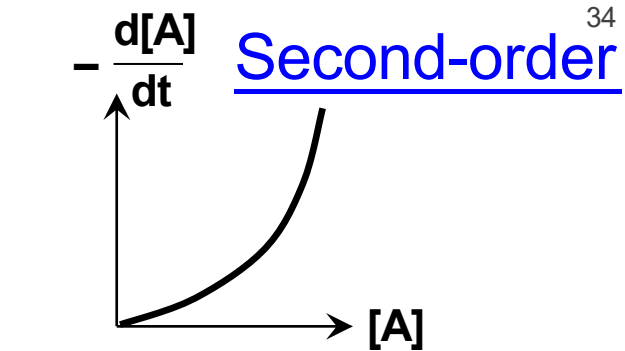
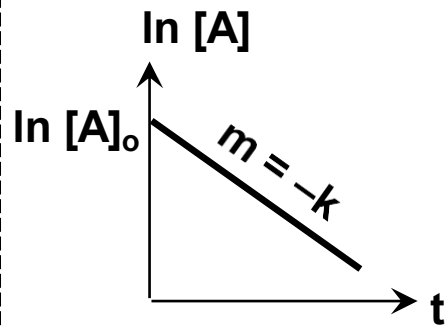
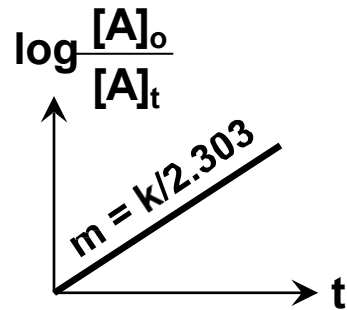
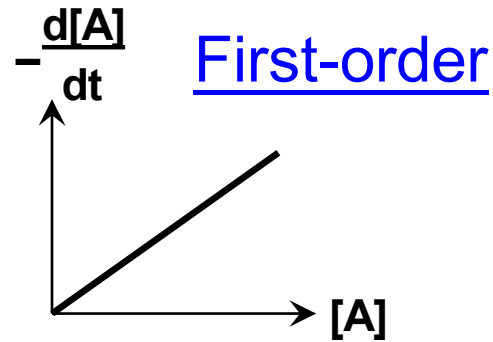
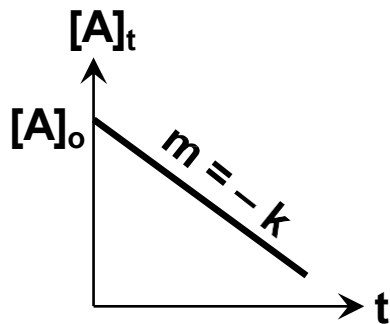
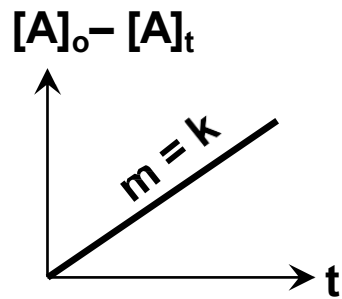
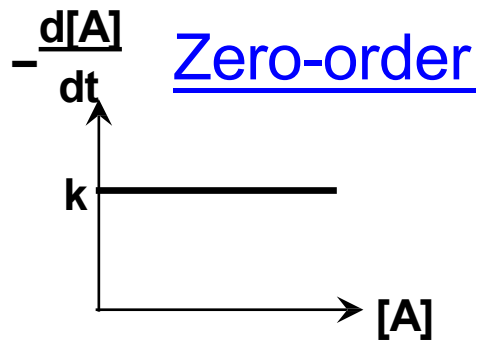
Characteristics of second order reaction:

$$\text{Second } t_{1/2} = 2 \times \text{First } t_{1/2}$$



Summary of Order of Reaction

Order of reaction	Rate Law	Integrated Rate Law	Half-life	Units of k
0	Rate = k	$[A]_o - [A]_t = kt$	$t_{1/2} = \frac{[A]_o}{2k}$	Concentration time ⁻¹ (Ms ⁻¹)
1	Rate = k[A]	$\ln \left(\frac{[A]_o}{[A]_t} \right) = kt$	$t_{1/2} = \frac{\ln 2}{k}$	time ⁻¹ (s ⁻¹)
2	Rate = k[A] ²	$\frac{1}{[A]_t} - \frac{1}{[A]_o} = kt$	$t_{1/2} = \frac{1}{k[A]_o}$	Concentration ⁻¹ time ⁻¹ (M ⁻¹ s ⁻¹)



Determining The Order of Reaction Involving a Single Reactant

1. Using Initial Rate method:

Example 1:

The data below is obtained for the reaction,



[P] (mol L ⁻¹)	8.5×10^{-3}	4.2×10^{-3}	2.0×10^{-3}
Initial rate (mol L ⁻¹ s ⁻¹)	2.4×10^{-6}	2.4×10^{-6}	2.4×10^{-6}

What is the order of reaction with respect to P?

Solution:

$$\text{Rate} = k [\text{P}]^x$$

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$$2.4 \times 10^{-6} = k (8.5 \times 10^{-3})^x \text{ ————— (1)}$$

$$2.4 \times 10^{-6} = k (4.2 \times 10^{-3})^x \text{ ————— (2)}$$

$$2.4 \times 10^{-6} = k (2.0 \times 10^{-3})^x \text{ ————— (3)}$$

To find value of x, use ratio of rate law from any two experiments.

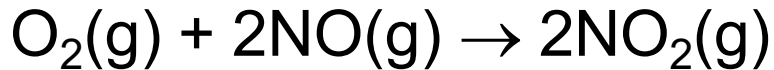
$$(1) \div (2) \rightarrow \frac{2.4 \times 10^{-6}}{2.4 \times 10^{-6}} = \left(\frac{8.5 \times 10^{-3}}{4.2 \times 10^{-3}} \right)^x$$

$$1 = 2^x \quad 2^0 = 2^x$$

$$\therefore x = 0$$

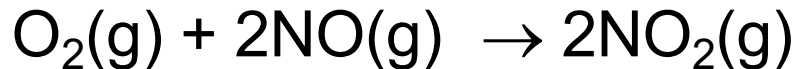
The reaction is zero-order with respect to P.

Example 2:



Experiment	<u>Initial Reactant Concentrations (molL⁻¹)</u>		<u>Initial Rate</u>
	O ₂	NO	(Ms ⁻¹)
1	1.10 x 10 ⁻²	1.30 x 10 ⁻²	3.21 x 10 ⁻³
2	2.20 x 10 ⁻²	1.30 x 10 ⁻²	6.40 x 10 ⁻³
3	1.10 x 10 ⁻²	2.60 x 10 ⁻²	12.8 x 10 ⁻³
4	3.30 x 10 ⁻²	1.30 x 10 ⁻²	9.60 x 10 ⁻³
5	1.10 x 10 ⁻²	3.90 x 10 ⁻²	28.8 x 10 ⁻³

Solution:



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$$\text{Rate} = k [\text{O}_2]^m [\text{NO}]^n$$

Compare 2 experiments in which the concentration of reactant varies while the other reactant(s) remains constant.

$$\frac{\text{Rate}_2}{\text{Rate}_1} = \frac{k[\text{O}_2]_2^m [\text{NO}]_2^n}{k[\text{O}_2]_1^m [\text{NO}]_1^n} = \frac{[\text{O}_2]_2^m}{[\text{O}_2]_1^m} = \left(\frac{[\text{O}_2]_2}{[\text{O}_2]_1} \right)^m$$

$$\frac{6.40 \times 10^{-3} \text{ Ms}^{-1}}{3.21 \times 10^{-3} \text{ Ms}^{-1}} = \left(\frac{2.20 \times 10^{-2} \text{ mol/L}}{1.10 \times 10^{-2} \text{ mol/L}} \right)^m$$

$$2 = 2^m \quad m = 1$$

The reaction is first-order with respect to O_2 .

Note : *Do a similar calculation for the other reactant(s).*

Continue:

To find the order with respect to NO, we compare experiment 3 and 1, in which $[O_2]$ is held constant and $[NO]$ is doubled:

$$\frac{\text{Rate 3}}{\text{Rate 1}} = \frac{k [O_2]_3^m [NO]_3^n}{k [O_2]_1^m [NO]_1^n} = \left(\frac{[NO]_3}{[NO]_1} \right)^n$$
$$\frac{12.8 \times 10^{-3} \text{ Ms}^{-1}}{3.21 \times 10^{-3} \text{ Ms}^{-1}} = \left(\frac{2.60 \times 10^{-2} \text{ mol/L}}{1.30 \times 10^{-2} \text{ mol/L}} \right)^n$$

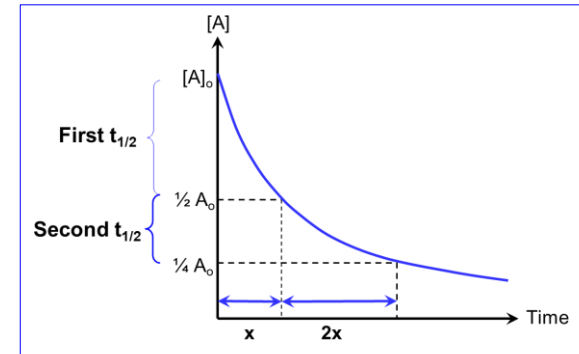
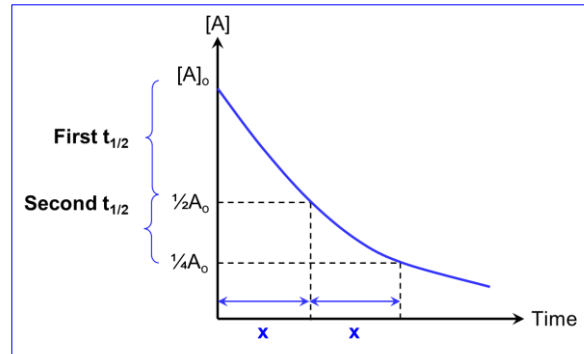
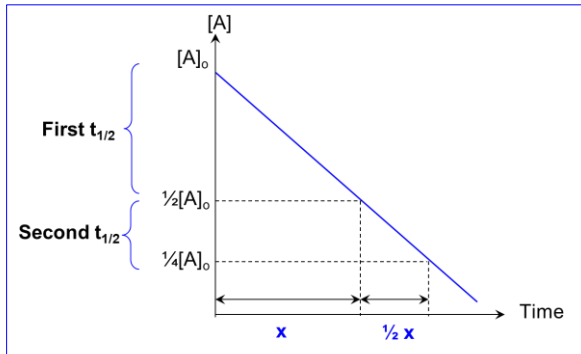
$$4 = 2^n \quad ; \quad n = 2$$

The reaction is second-order with respect to NO.

Thus, the rate law is : $\text{Rate} = k[O_2][NO]^2$

2. Using Half-life based on graph concentration against time:

The half-life values from the plot of **concentration versus time** can be used to deduce the zero- , first- or second-order reactions:



First $t_{1/2} = 2 \times \text{Second } t_{1/2}$

First $t_{1/2} = \text{Second } t_{1/2}$

First $t_{1/2} = \frac{1}{2} \text{Second } t_{1/2}$

Example:

The following results were obtained from an experimental investigation on dissociation of dinitrogen pentoxide at 45°C.



T(min)	0	10	20	30	40	50	60
[N ₂ O ₅] x 10 ⁻⁴ M	176	124	93	71	53	39	29

Plot graph of [N₂O₅] vs time, determine the order of the reaction.

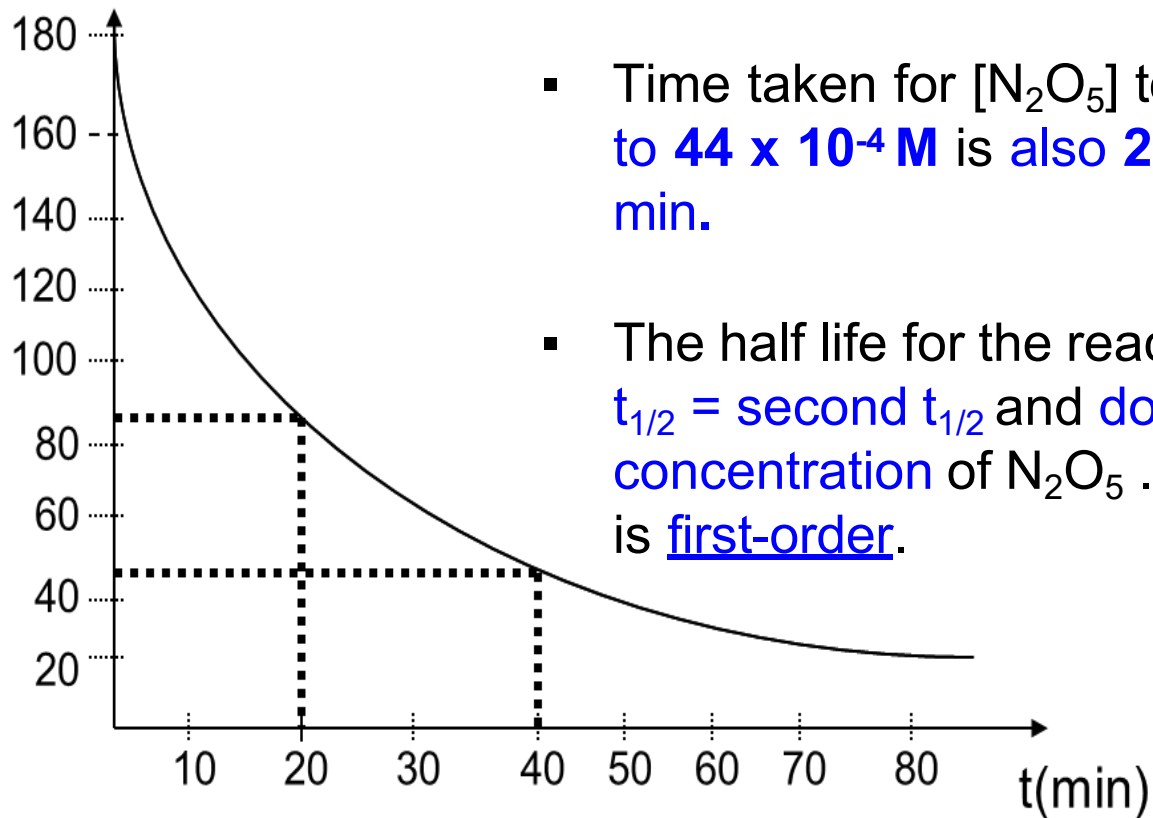
Solution:

Based on the graph:

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- Time taken for $[N_2O_5]$ to change from $176 \times 10^{-4} \text{ M}$ to $88 \times 10^{-4} \text{ M}$ is **20 min** @ First $t_{1/2} = 20 \text{ min}$.
- Time taken for $[N_2O_5]$ to change from $88 \times 10^{-4} \text{ M}$ to $44 \times 10^{-4} \text{ M}$ is **also 20 min** @ Second $t_{1/2} = 20 \text{ min}$.
- The half life for the reaction is a **constant** @ **first $t_{1/2} = \text{second } t_{1/2}$** and **does not** depend on the initial **concentration** of N_2O_5 . Thus, the order of reaction is first-order.

$[N_2O_5] \times 10^{-4} \text{ M}$



3. Using Linear Graph method based on the integrated rate equation and rate law:

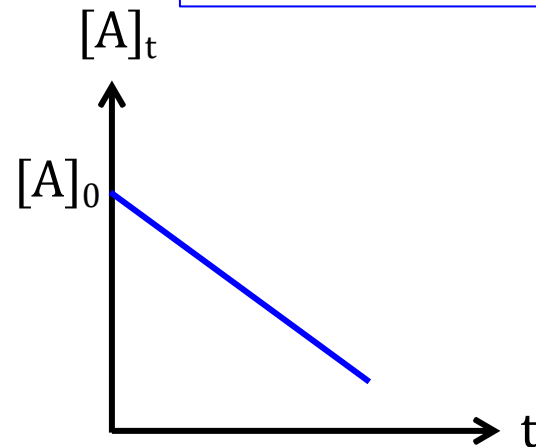
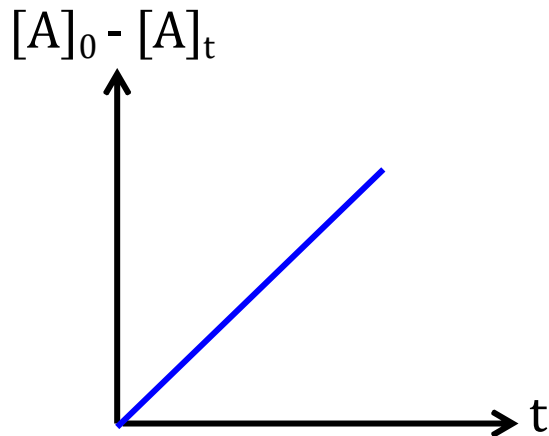
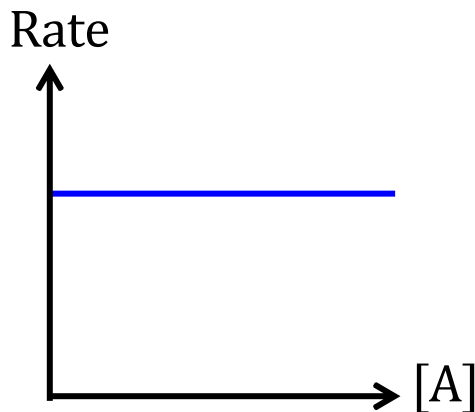
The order of the reaction can be determined from the linear graph method **based on the rate law and integrated rate equation.**

Zero order

$A \rightarrow \text{product}$

$$\text{Rate} = k$$

$$[A]_0 - [A]_t = kt$$



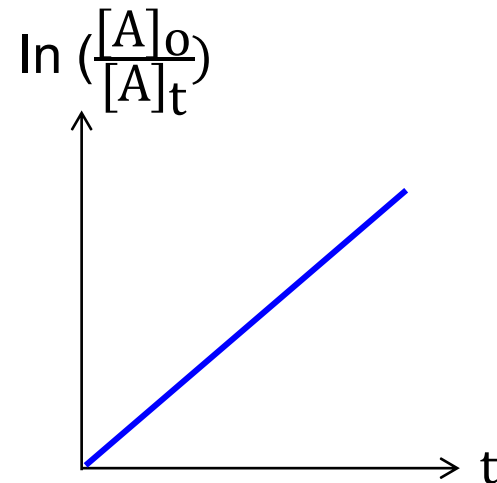
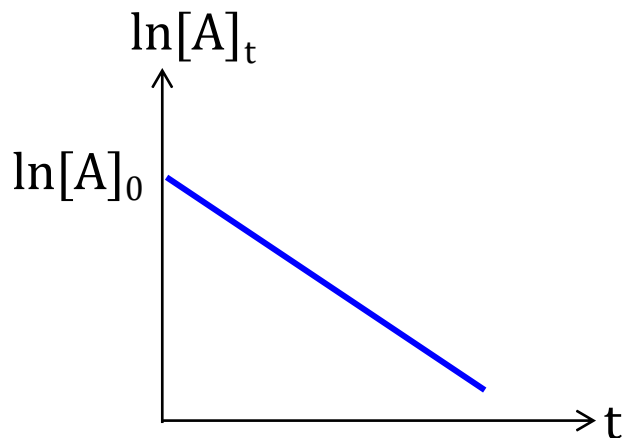
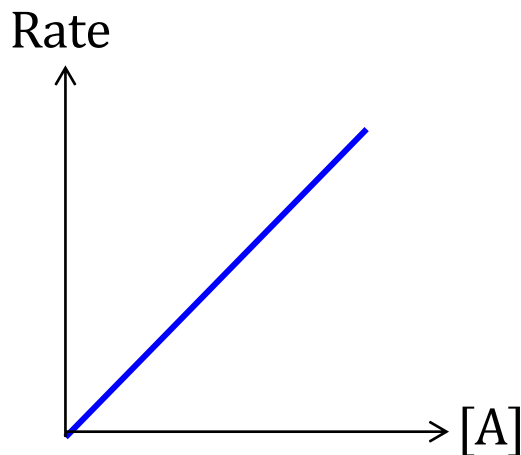
3. Using Linear Graph method based on the integrated rate equation and rate law:

First order

$A \rightarrow \text{product}$

$$\text{Rate} = k [A]^1$$

$$\ln \left(\frac{[A]_0}{[A]_t} \right) = kt$$



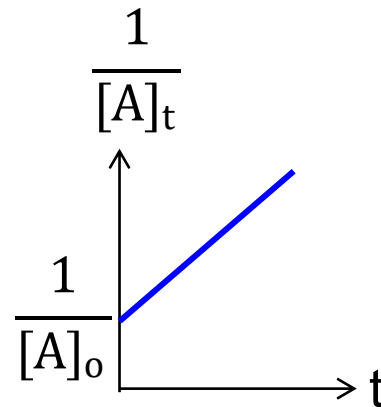
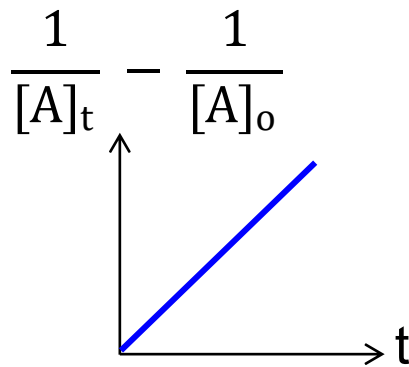
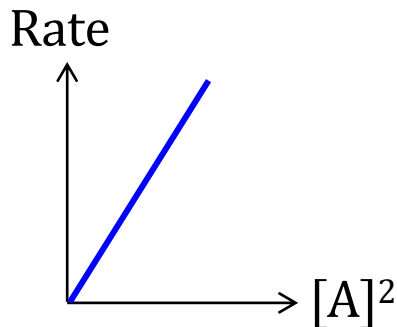
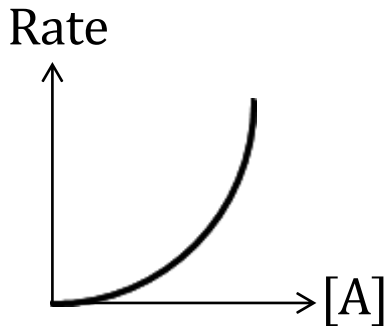
3. Using Linear Graph method based on the integrated rate equation and rate law:

Second order

$A \rightarrow \text{product}$

$$\text{Rate} = k[A]^2$$

$$\frac{1}{[A]_t} - \frac{1}{[A]_o} = kt$$



Example:

Dinitrogen pentoxide, N_2O_5 , is not stable. In the gas phase or when dissolved in a non-aqueous solvent, like carbon tetrachloride, it decomposes by a first-order reaction into N_2O_4 and O_2 .



The rate law is : $\text{rate} = k [\text{N}_2\text{O}_5]$

At 45°C , the rate constant for the reaction in carbon tetrachloride is $6.22 \times 10^{-4} \text{ s}^{-1}$. If the initial concentration of N_2O_5 in the solution is 0.100 M , how many seconds will it take for the concentration to drop to 0.010 M ?

Solution:

$$[\text{N}_2\text{O}_5]_o = 0.100 \text{ M}$$

$$k = 6.22 \times 10^{-4} \text{ s}^{-1}$$

$$[\text{N}_2\text{O}_5]_t = 0.010 \text{ M}$$

$$t = ?$$

$$\ln \left(\frac{[\text{N}_2\text{O}_5]_o}{[\text{N}_2\text{O}_5]_t} \right) = kt$$

$$\ln \left(\frac{0.100}{0.010} \right) = 6.22 \times 10^{-4} \times t$$

$$t = \mathbf{3.70 \times 10^3 \text{ s}}$$

Example 1:

The decomposition of formic acid, HCOOH , in the gaseous phase at elevated temperature,

$$\text{HCOOH(g)} \rightarrow \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$$

is first order with a rate constant of $1.0 \times 10^{-2} \text{ s}^{-1}$. Write the rate equation of this reaction. Calculate the time that has elapsed if the mass of formic acid has decreased to 30% of its original value.

Solution:

Rate law: **Rate = k [HCOOH]**

Initial concentration = $[\text{HCOOH}]_0$

$$[\text{HCOOH}]_t = \frac{30}{100} [\text{HCOOH}]_0$$

$$\ln \frac{[\text{HCOOH}]_0}{[\text{HCOOH}]_t} = kt$$

$$\ln \frac{[\text{HCOOH}]_0}{\frac{30}{100} [\text{HCOOH}]_0} = 1.0 \times 10^{-2} \times t$$

$$\ln \frac{100}{30} = 1.0 \times 10^{-2} \times t \quad = \mathbf{120.40s}$$

Example 2:

Determine the percent H_2O_2 that decomposes using $k = 6.40 \times 10^{-5} \text{ s}^{-1}$ if the time for the concentration to decompose is 600.0 s after the reaction begins.

Solution:

The reaction is **first-order**.

Initial concentration = $[\text{H}_2\text{O}_2]_0$

$$\ln \frac{[\text{H}_2\text{O}_2]_0}{[\text{H}_2\text{O}_2]_t} = kt$$

$$\ln \frac{[\text{H}_2\text{O}_2]_0}{[\text{H}_2\text{O}_2]_t} = (6.4 \times 10^{-5})(600)$$

$$\frac{[\text{H}_2\text{O}_2]_0}{[\text{H}_2\text{O}_2]_t} = 1.039$$

$$\frac{[\text{H}_2\text{O}_2]_t}{[\text{H}_2\text{O}_2]_0} = 0.962$$

$$\begin{aligned} \% \text{H}_2\text{O}_2 \text{ left} &= 0.962 \times 100 \\ &= 96.23\% \end{aligned}$$

$$\begin{aligned} \% \text{ decomposed} &= 100 - 96.2 \\ &= 3.77\% \end{aligned}$$



1.2 Collision Theory

COLLISION THEORY

- According to the collision theory, chemical reactions occur as a result of collisions between reacting molecules. **HOWEVER...**not all collisions result in a chemical reaction.
- When the frequency of collisions per second is compared with the rate of reaction, only a small fraction of the collisions that occur results in a reaction

$$\text{Rate} \propto \frac{\text{Number of effective collisions}}{\text{time}}$$

COLLISION THEORY

- According to the Collision Theory, there are three conditions for a chemical reaction to occur:
 - 1) Molecules of reactants **must collide to react**.
 - 2) The reacting molecules must collide in a **correct orientation**.
 - 3) The colliding molecules must possess a **minimum** amount of **energy**, known as **activation energy**, to initiate a chemical reaction.

COLLISION THEORY

- ❑ Collisions that produce a chemical reaction are called effective collisions.
- ❑ There are two conditions for an effective collision to occur:
 - 1) Collision must be in the correct orientation in order to form products.
 - 2) The colliding molecules must possess energy which is equal or greater than activation energy.

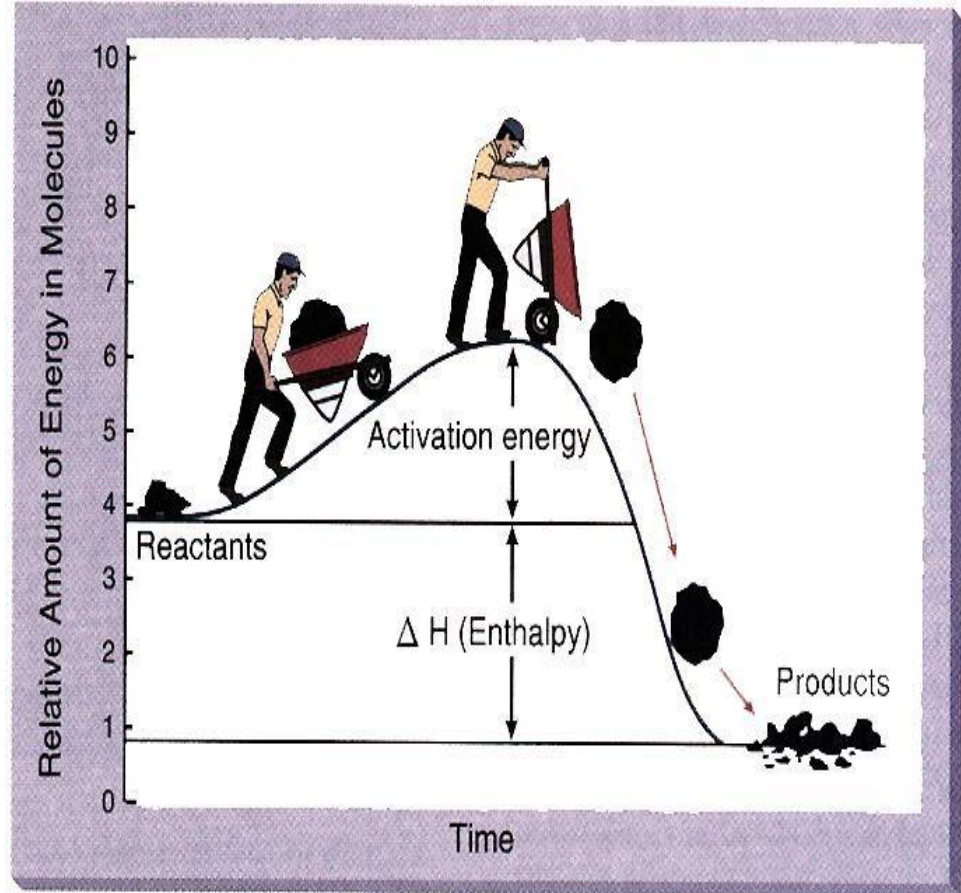
COLLISION THEORY

Activation energy (E_a)

The **minimum energy** required to initiate a chemical reaction

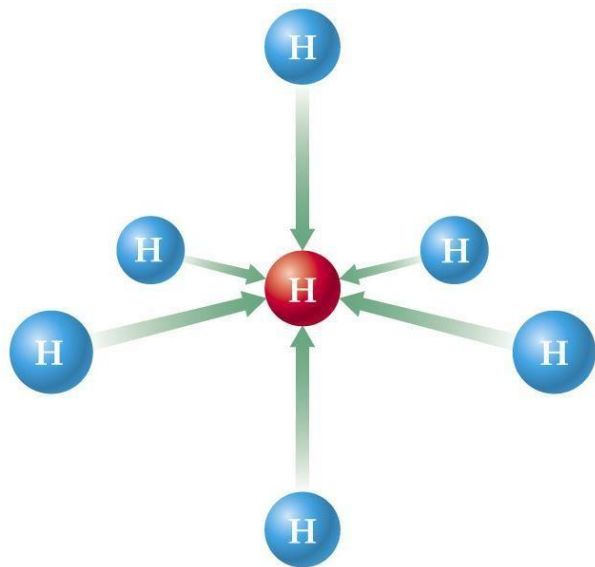
or

The **minimum amount** of energy needed to activate or energize molecules so they can undergo a chemical reaction.

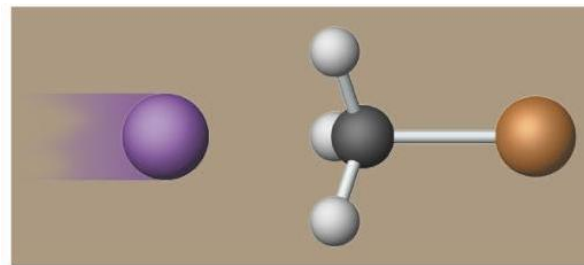


COLLISION THEORY

Examples of the importance of a correct collision orientation.

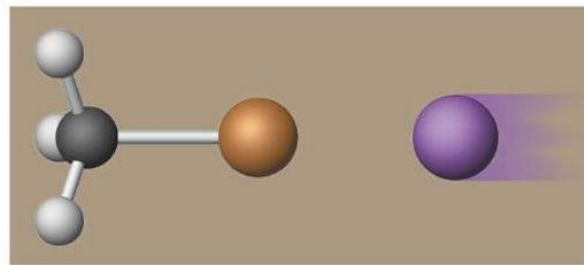


Orientation is unimportant



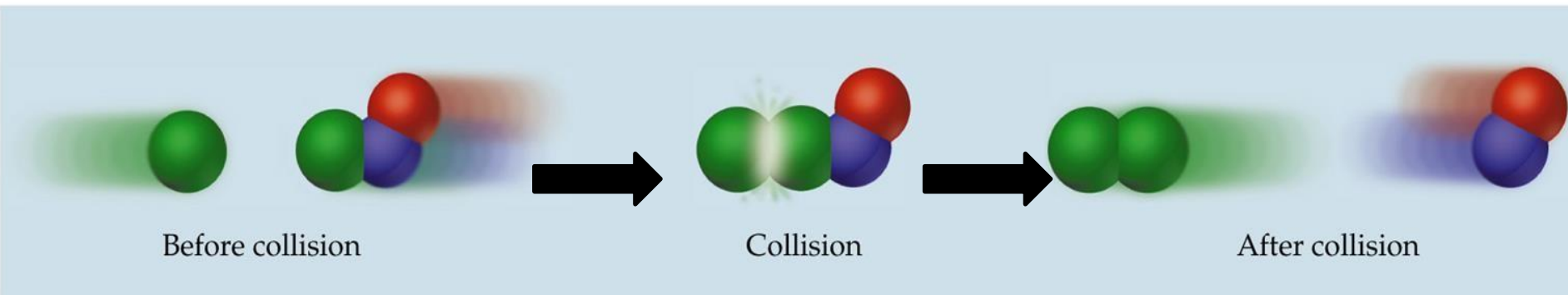
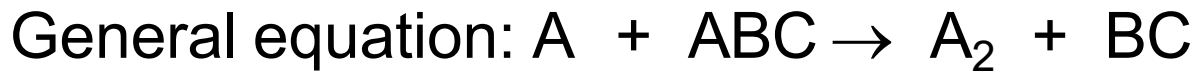
(a)

Correct
orientation



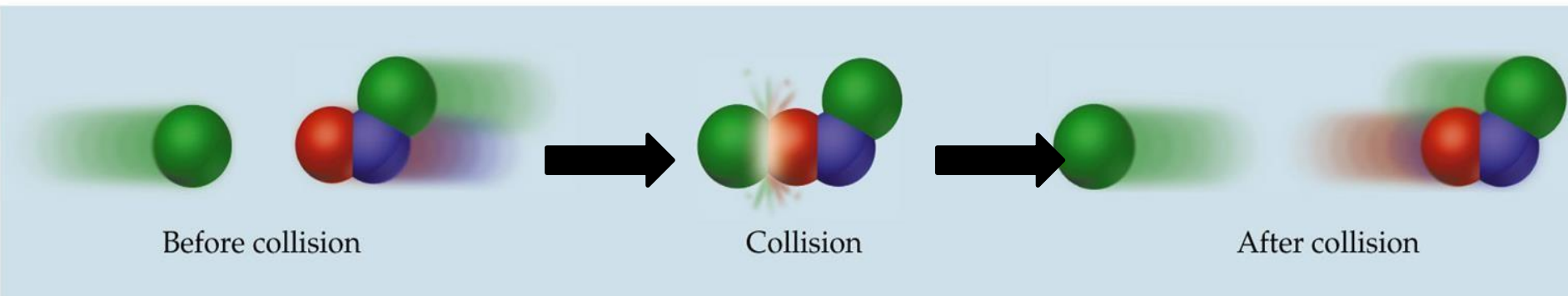
(b)

Incorrect
orientation



(a) Effective collision

Effective collision



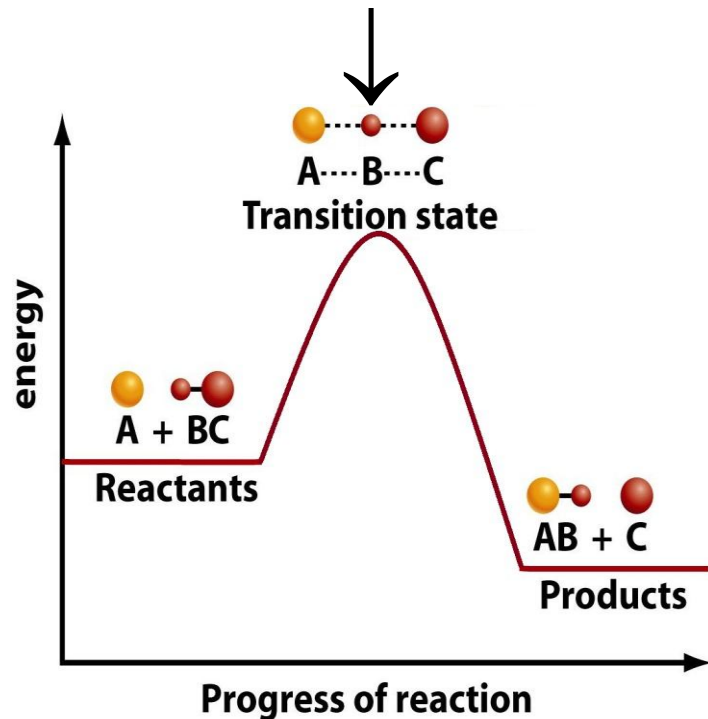
Ineffective collision

Transition State Theory

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Activated complex

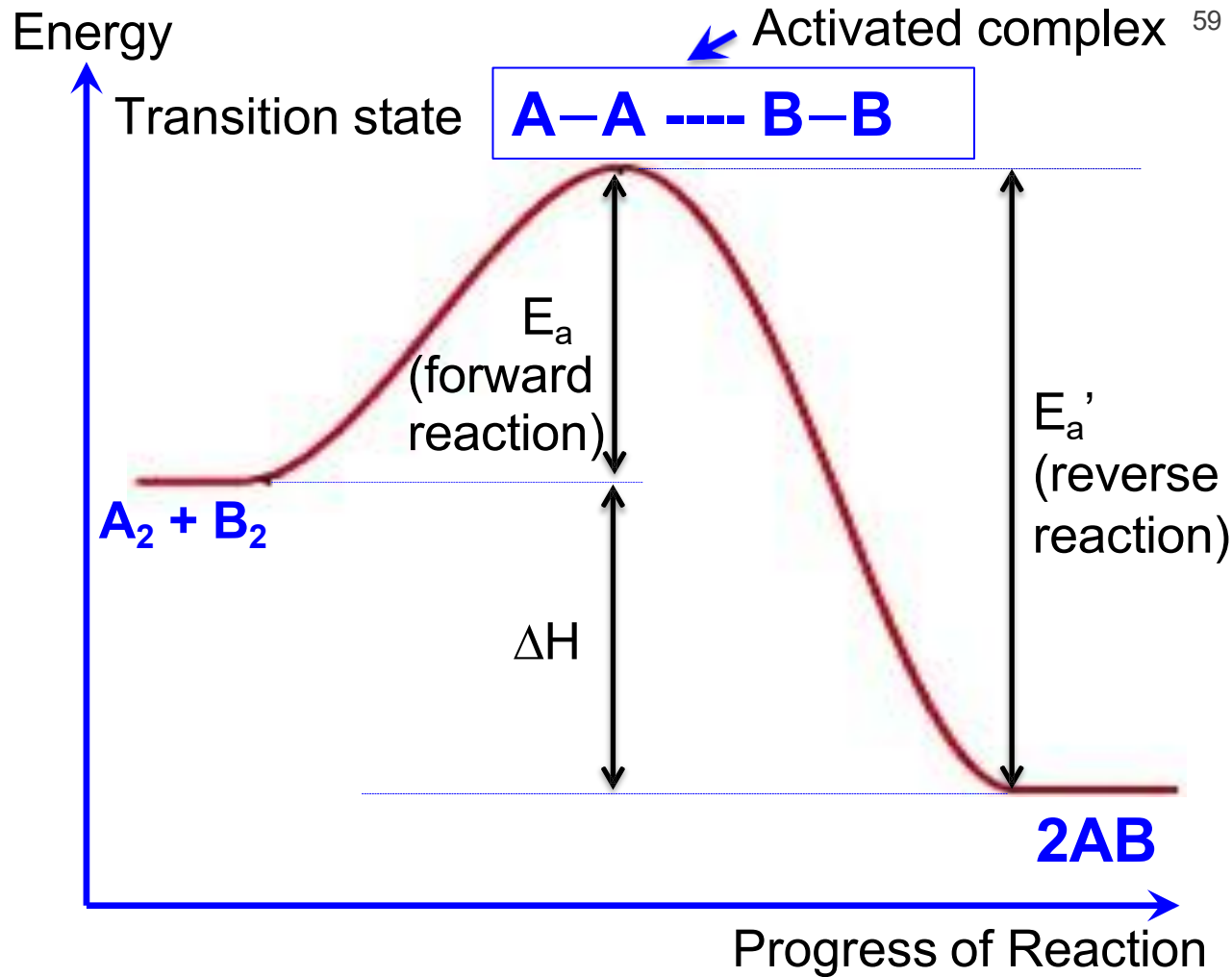
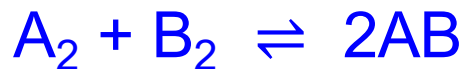
- Reactants form an activated complex or transition state with maximum potential energy when approach each other.
- The transition state is an **unstable** transitory combination of reactant molecules.
- The combination can either go on to form products or fall apart to return to the unchanged reactants.
- The energy difference between the reactants and the maximum potential energy is referred to as the activation energy.



Energy Profile Diagram of a reaction

- ❑ Shows potential energy plotted as a function of the progress of the reaction.
- ❑ The difference in potential energies between products and reactants is ΔH .
- ❑ Reactant molecules must have enough energy to overcome an energy barrier, the E_a , which separating products from reactants.

Exothermic Reaction:



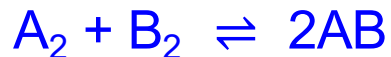
Activated complex

Transition state

 E_a'
(reverse
reaction) ΔH E_a (forward
reaction)

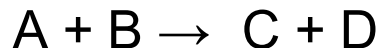
Progress of Reaction

Energy

Endothermic Reaction:

Example:

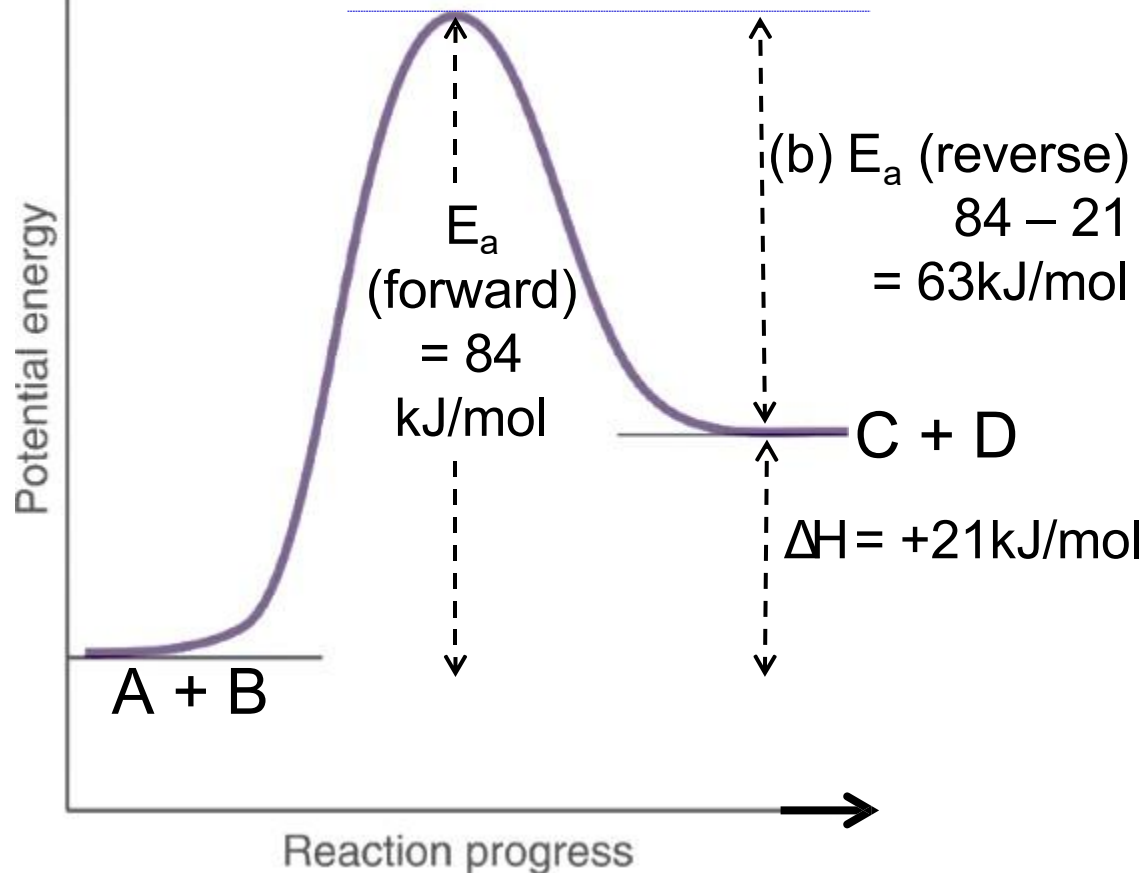
For the following reaction:



the enthalpy change of the forward reaction is +21 kJ/mol. The activation energy of the forward reaction is 84 kJ/mol.

- Sketch the reaction profile.
- What is activation energy of the reverse reaction?

Solution:





1.3

Factors

Affecting

Reaction Rate

Factors Affecting Rate of Reaction

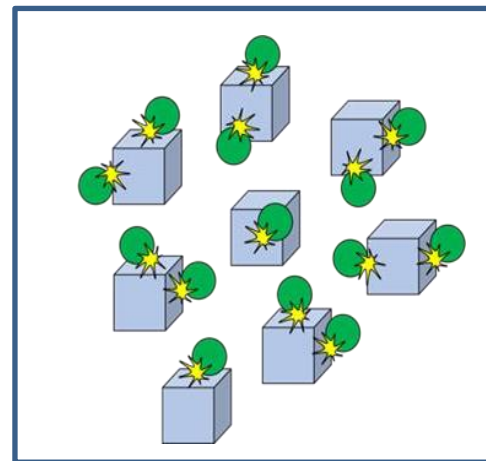
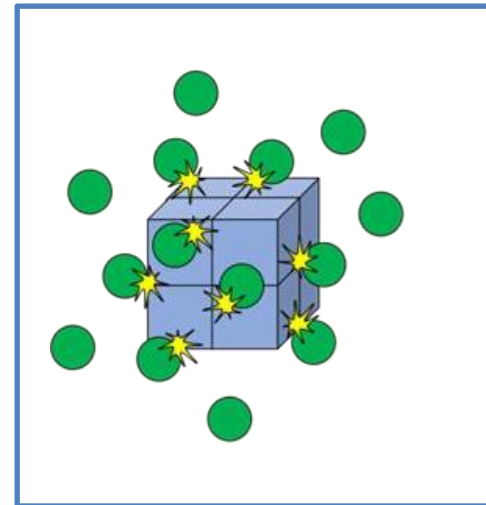
- In general rate of reaction can be increased by increasing probability of effective collision, either by increasing frequency of collision or lowering the activation energy.
- Factors that affect rate of reaction are:
 - 1) **Particle size of reactants**
 - 2) **Concentrations or pressure**
 - 3) **Temperature**
 - 4) **Catalyst**

1. Particle size

The **smaller the size** of reacting particles,

- its surface area gets larger.
- when surface area increase, the number of collision increases,
- thus, increasing the frequency of effective collisions

Hence, rate of reaction increases.



2. Concentrations of reactants

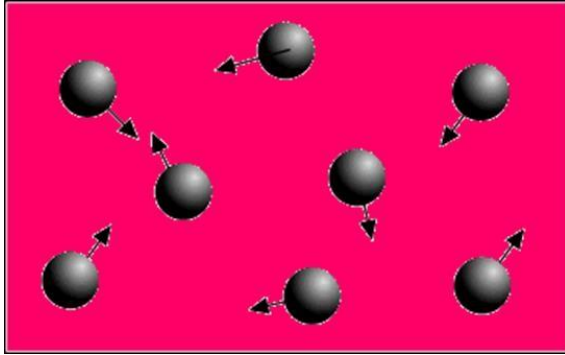
The **greater the concentration** of reactants,

- the number of reactant particles per unit volume **increases**, thus particles are **closer** together,
- frequency of collision **increases**,
- frequency of effective collision **increases**.

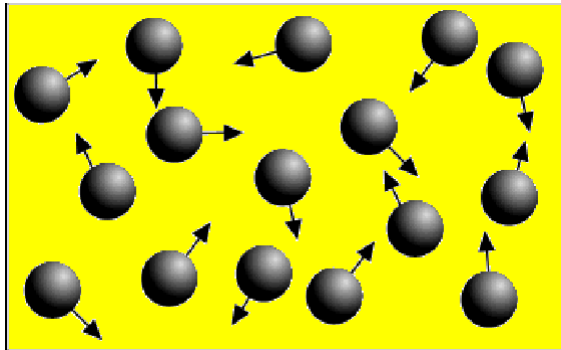
Thus, rate of reaction **increases**.

Note: (Similar explanation for increase in pressure)

Concentrations of reactants



Low conc. = fewer collisions



High conc. = greater collisions

Increasing pressure



Low pressure = fewer collisions



High pressure = greater collisions

3. Temperature

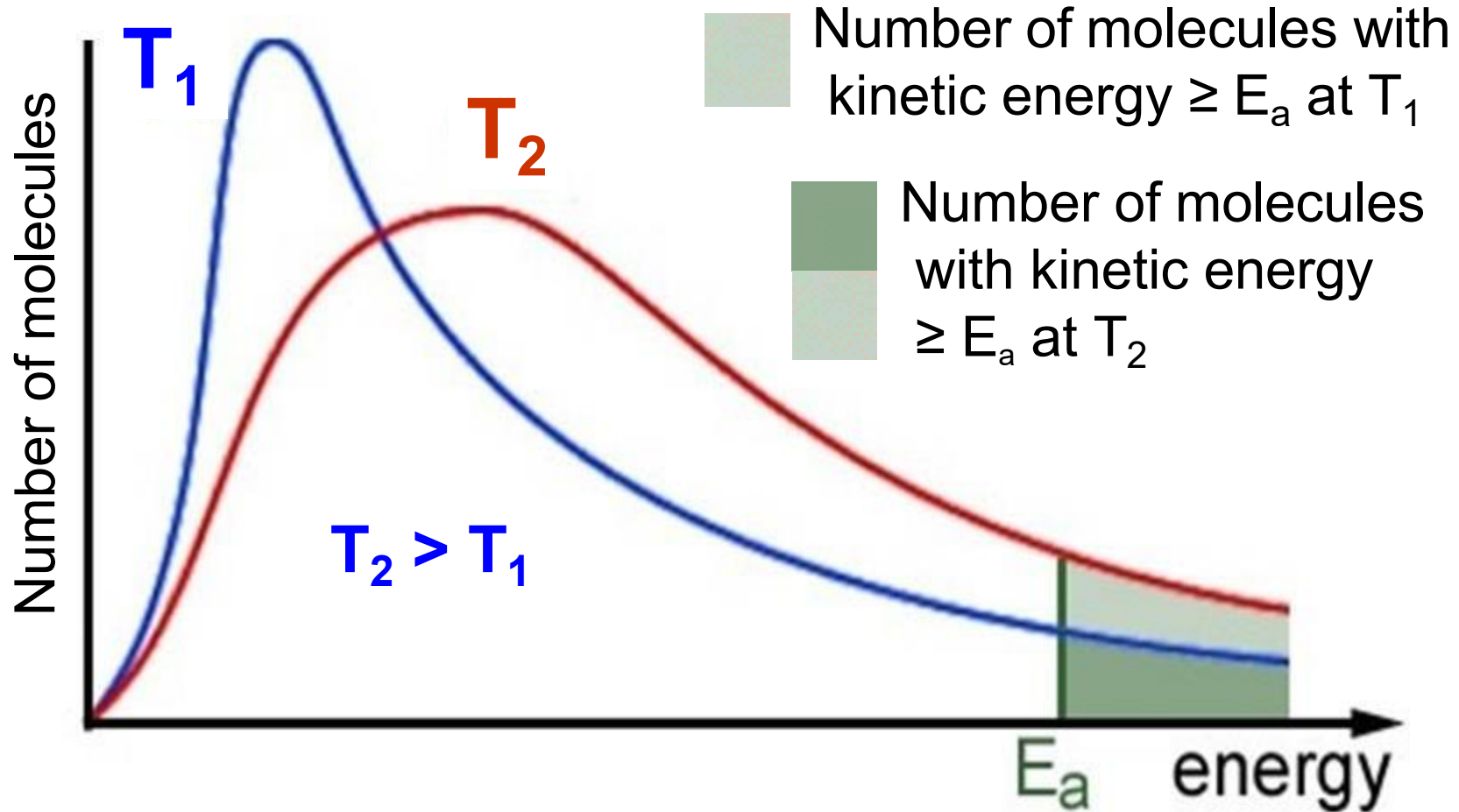
As temperature **increases**,

- molecules move **faster** as average kinetic energy of the molecules **increases**,
- number of molecules which possess energy **equal or greater** than the activation energy **increases**,
- frequency of effective collisions **increase**,

Hence, rate of reaction **increases**.

Note: Change in temperature also affect the rate constant, k .

Maxwell–Boltzmann Distribution Curve



Maxwell–Boltzmann Distribution Curve

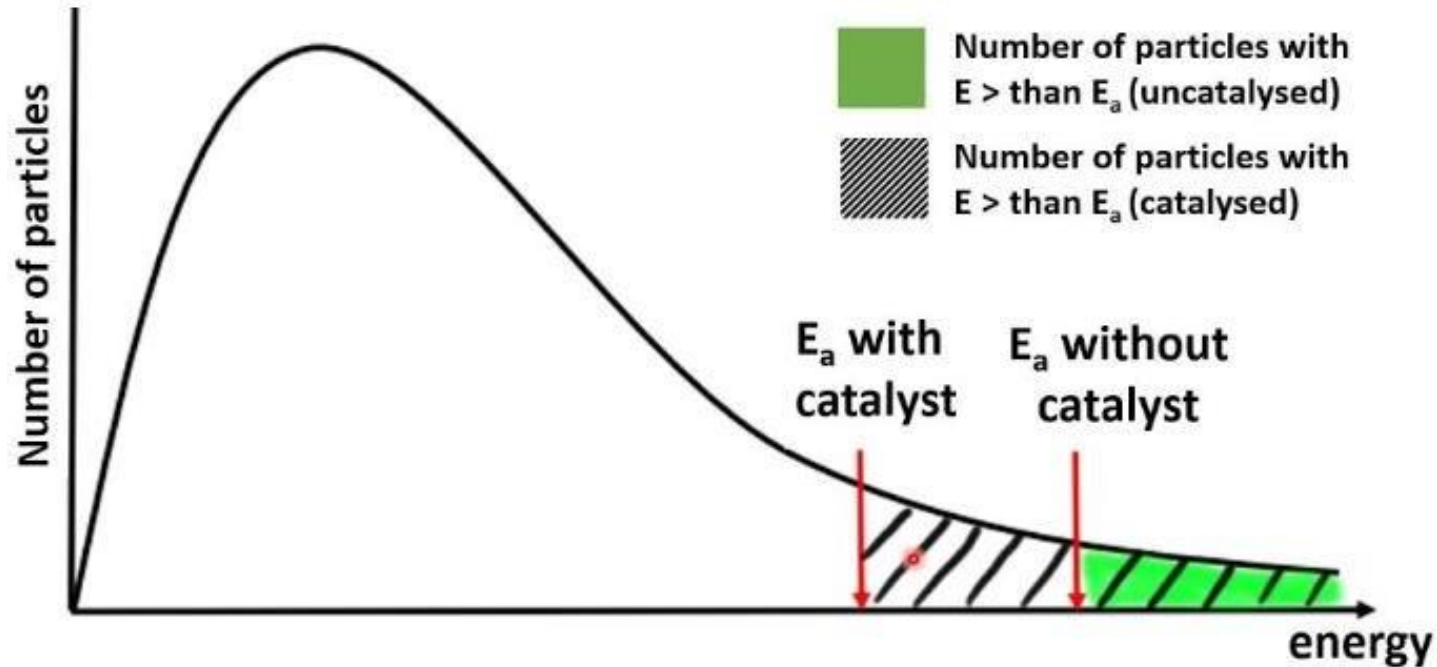
- The diagram shows a distribution curve for a gas at temperature T_1 and T_2 .
- The shaded areas represent the number of molecules with kinetic energy equal or greater than the activation energy, E_a .
- When temperature increases, the average kinetic energy of molecules increases.
- The number of reactant molecules that possess the energy equal to or greater than the activation energy, E_a increases.
- The frequency of effective collisions increases thus results in higher reaction.

Hence, the rate of reaction increases.

4. Addition of catalyst

- Catalyst is a substance that **increases the rate** of a chemical reaction without itself being consumed.
- A catalyst provides an alternative pathway with lower activation energy for the reaction.
- This means that the number of molecules with energy higher than the activation energy **increases**.
- The number of effective collision **increases** thus, rate of reaction **increases**.

4. Addition of catalyst

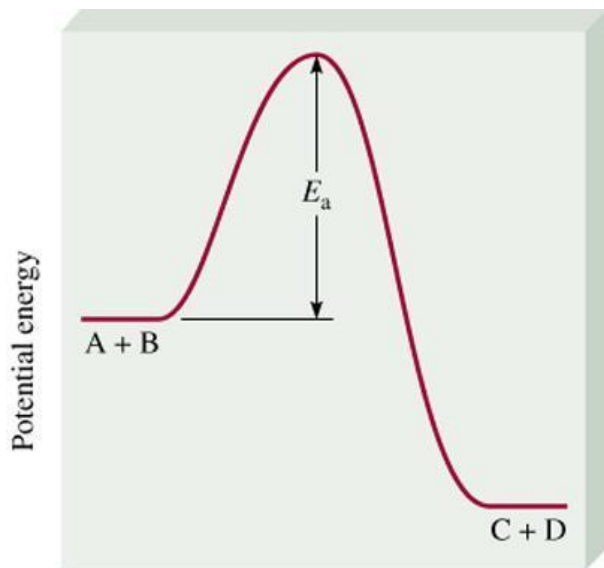


In the catalysed reaction, a greater number of molecules having energy greater than the activation energy, therefore the rate of reaction increases.

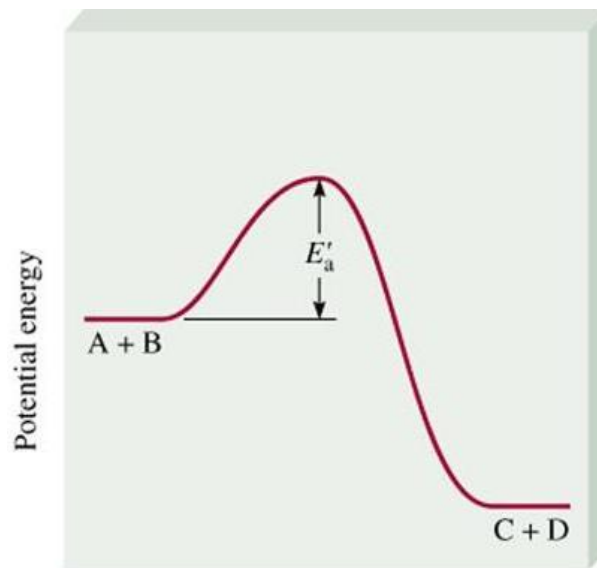
The Energy Profile Diagram for a Catalysed Reaction vs Uncatalysed Reaction

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Reaction_{without catalyst}

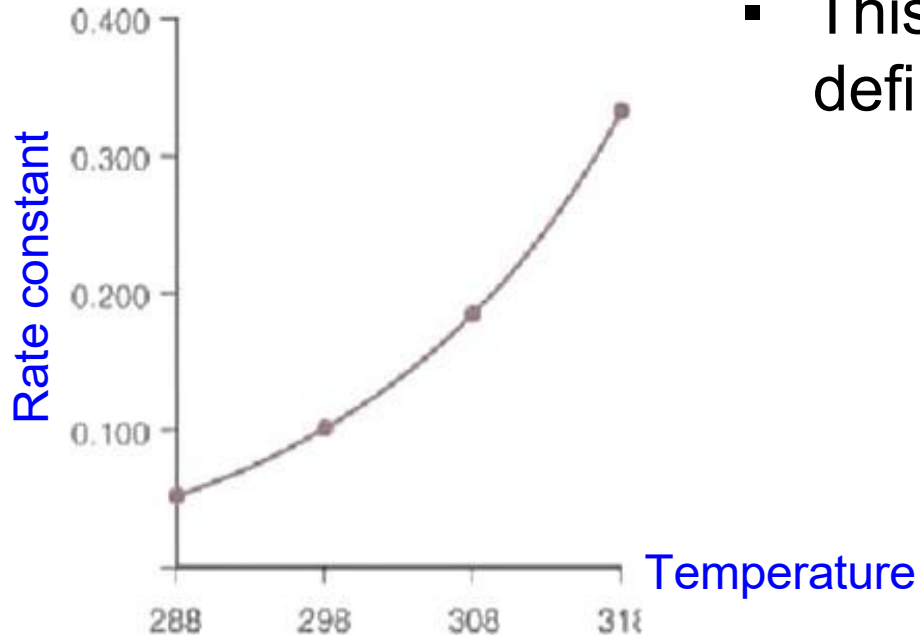


Reaction_{with catalyst}



Arrhenius Equation

- A plot of **rate constant versus temperature** shows an exponential increase.



- This relationship between k and T is defined by the Arrhenius equation

$$k = A e^{-E_a/RT}$$

A = collision frequency factor

E_a = activation energy

R = molar gas constant ($8.314 \text{ Jmol}^{-1}\text{K}^{-1}$)

T = absolute temperature

- The Arrhenius equation can be written in a non-exponential form that is often more convenient to use. Taking the natural logarithm of both sides of the Arrhenius equation,

$$\ln k = -\frac{E_a}{R} \left(\frac{1}{T} \right) + \ln A$$

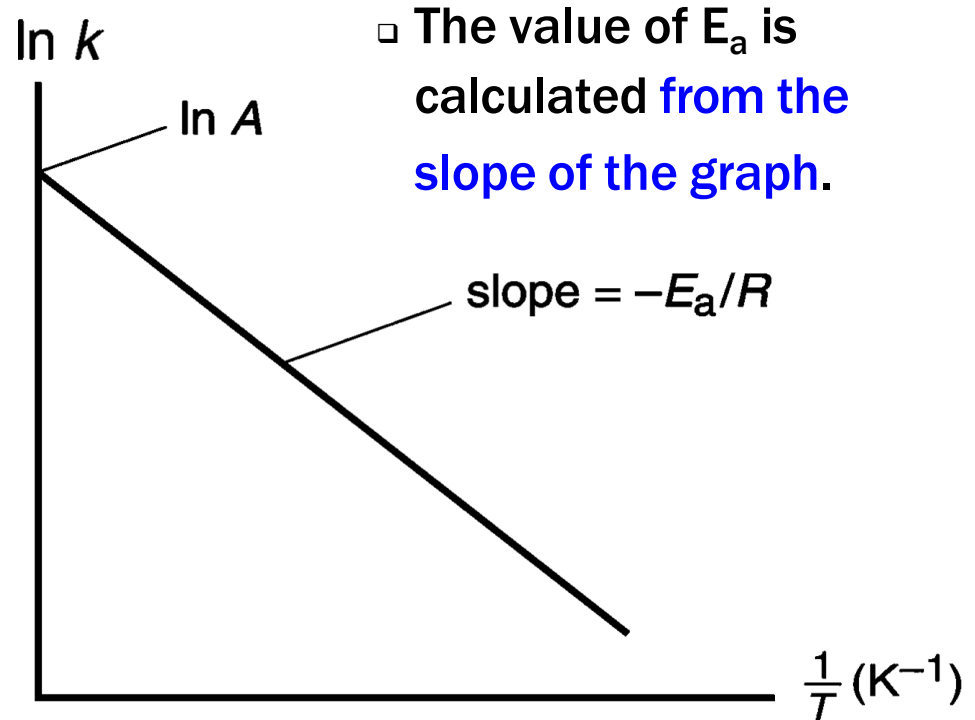
- Taking two measured values of the rate constant, k , at two different temperatures, the Arrhenius equation is written as;

$$\ln \left(\frac{k_1}{k_2} \right) = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

By plotting a linear graph of $\ln k$ vs $1/T$, the value of E_a can be determined graphically.

$$\ln k = -\frac{E_a}{R} \left(\frac{1}{T} \right) + \ln A$$

$\downarrow$ $\downarrow$ $\downarrow$ $\swarrow$
 y m x $+ c$



- Based on the Arrhenius equation, catalyst acts by reducing the E_a , thus increasing the rate constant, k .

$$\ln k = - \frac{E_a}{R} \left(\frac{1}{T} \right) + \ln A$$

- Since k is directly proportional to the rate of reaction, therefore, increasing k will increase the rate of reaction.
- In the presence of catalyst,
 E_a decreases, k increases, thus rate of reaction increases.

Relationship Between Temperature to the Rate Constant, k

$$k = A e^{-E_a/RT}$$

- Increasing **temperature** causes the term $e^{-\frac{E_a}{RT}}$ to be **larger**, thus **increasing** the k value.

Thus, the rate of reaction **increases**.

- Increasing E_a causes the term $e^{-\frac{E_a}{RT}}$ to be smaller, thus decreasing k

Relationship Between Activation Energy to the Rate Constant, k

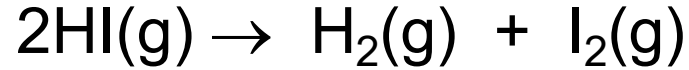
$$k = A e^{-E_a/RT}$$

- Increasing E_a causes the term $e^{-\frac{E_a}{RT}}$ to be **smaller**,
- thus **decreasing** the k value.

Thus, the rate of reaction **decreases**.

Example:

The decomposition of hydrogen iodide,



has rate constants of $9.51 \times 10^{-9} \text{ L mol}^{-1} \text{ s}^{-1}$ at 500 K and $1.10 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$ at 600 K. Find E_a .

Solution:

$$\ln \left(\frac{k_1}{k_2} \right) = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln \left(\frac{9.51 \times 10^{-9}}{1.10 \times 10^{-5}} \right) = \frac{E_a}{8.314} \left(\frac{1}{600} - \frac{1}{500} \right)$$

$$= 1.76 \times 10^5 \text{ J/mol} = 176 \text{ kJ/mol}$$

The slide features abstract geometric designs in the corners. The top-left corner is filled with overlapping triangles in shades of blue, green, and red. The bottom-right corner contains a cluster of overlapping triangles in various shades of gray.

THE END